US006256999B1

(12) **United States Patent**
Chase

(10) **Patent No.: US 6,256,999 B1**
(45) **Date of Patent: Jul. 10, 2001**

(54) **TEMPERATURE CONTROL OF HIGH TEMPERATURE SUPERCONDUCTING THIN FILM FILTER SUBSYSTEMS**

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(73) Assignee: **Superconductor Technologies, Inc.**, Santa Barbara, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **09/634,695**

(22) Filed: **Aug. 8, 2000**

Related U.S. Application Data

(63) Continuation of application No. 09/204,897, filed on Dec. 3, 1998, now Pat. No. 6,098,409.

(51) **Int. Cl.**[7] **F25B 9/00**
(52) **U.S. Cl.** **62/6**
(58) **Field of Search** 62/6, 908

(56) **References Cited**

U.S. PATENT DOCUMENTS

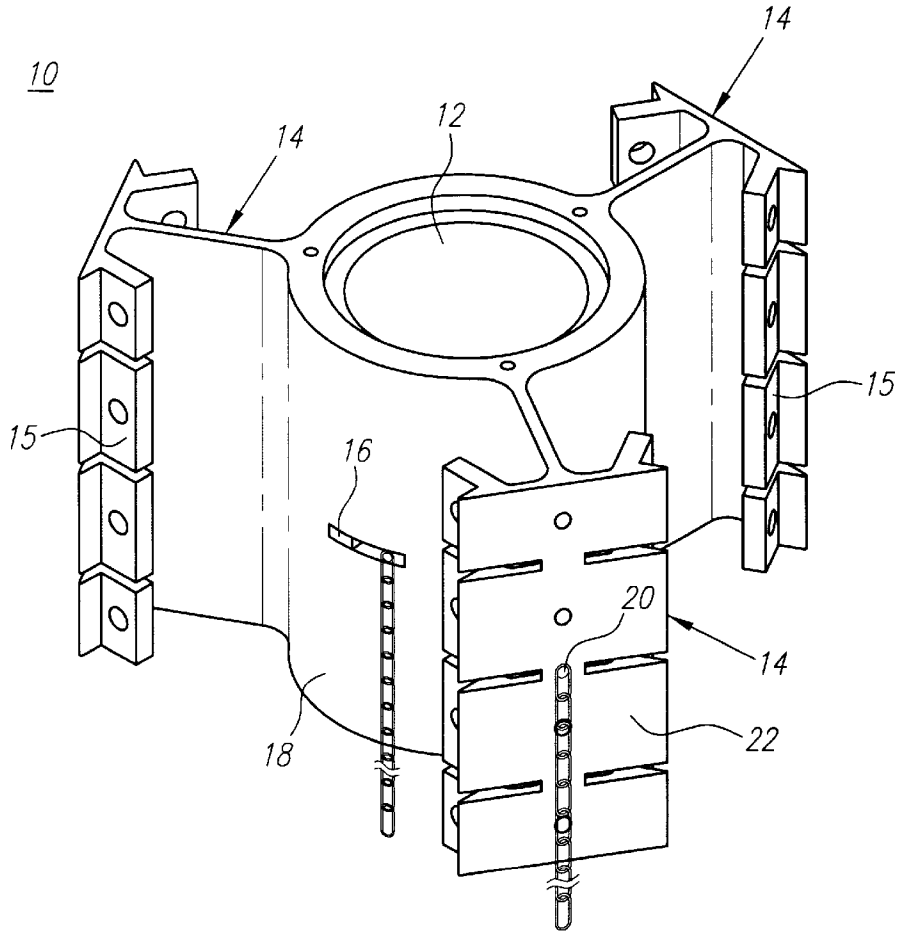
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5,535,593	7/1996	Wu et al.	62/6
5,813,235	9/1998	Peterson	62/6

Primary Examiner—William Doerrler
Assistant Examiner—Malik N. Drake
(74) *Attorney, Agent, or Firm*—Lyon & Lyon LLP

(57) **ABSTRACT**

Systems and methods for regulating, for example, the temperature of high temperature superconducting (HTSC) thin film filters. A multi-stage temperature controller comprises a first loop and a second loop. The first loop is used to regulate the cold finger temperature of, for example, a Stirling cycle cryocooler, and the second loop is used to set a reference for the first loop based upon a comparison between a reference signal and a signal received from a cold stage temperature sensor. A filter loop also may be used to regulate the temperature of independent HTSC thin film filter carrier packages.

9 Claims, 25 Drawing Sheets



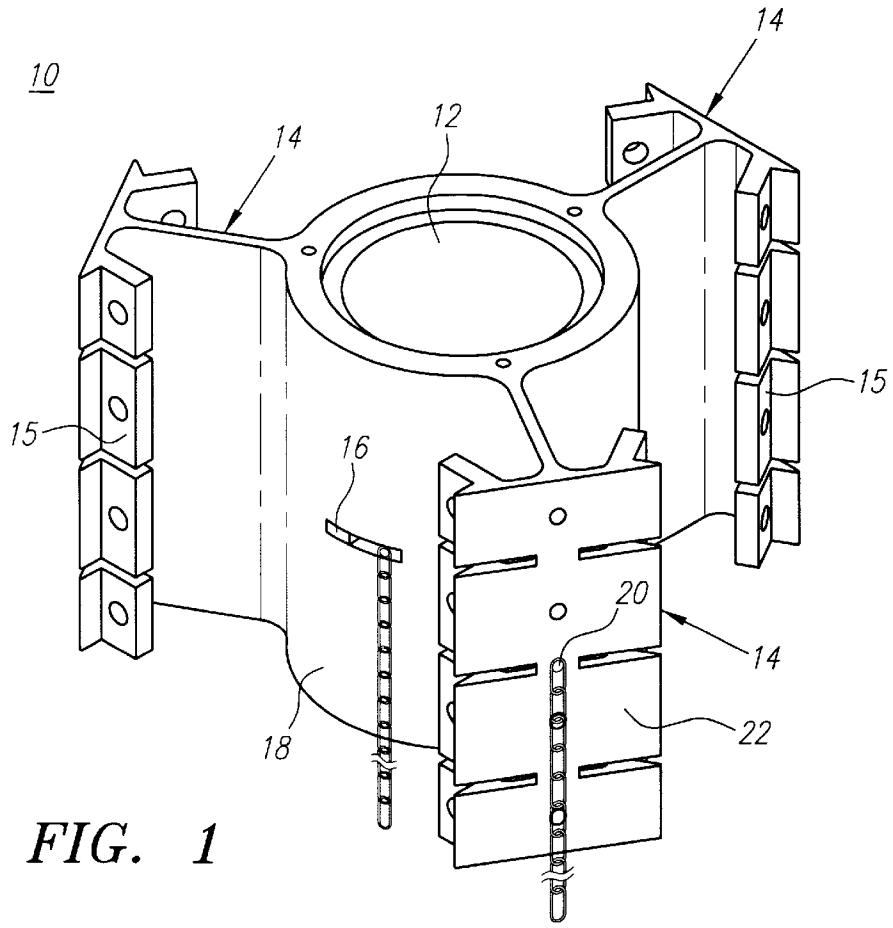

10
14
12
14
15
15
16
20
14
18
22

FIG. 1

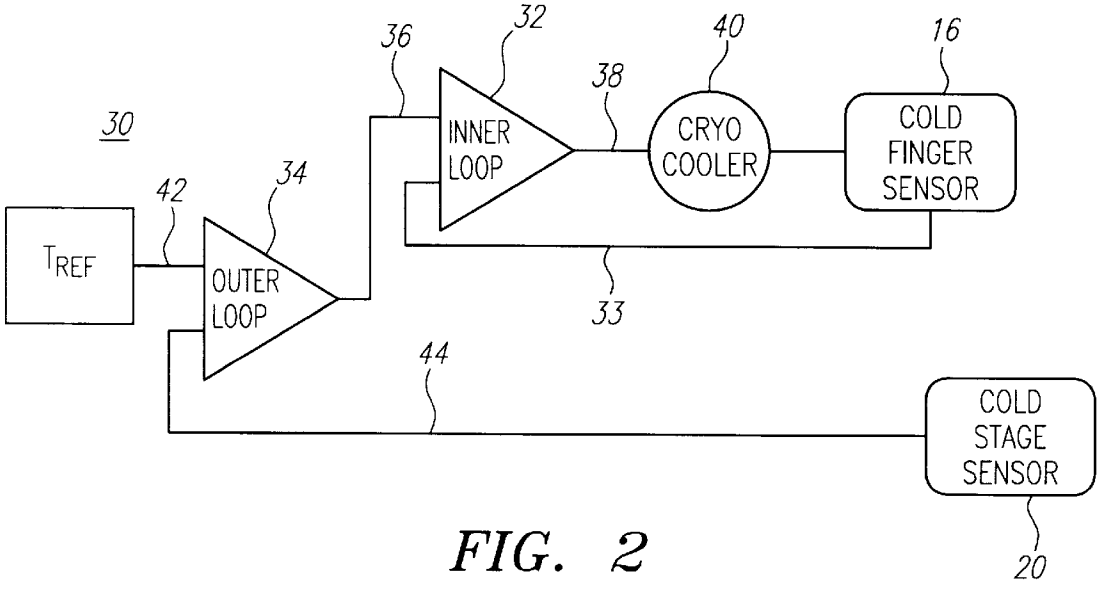

30
36
32
40
16
38
INNER LOOP
CRYO COOLER
COLD FINGER SENSOR
42
34
TREF
OUTER LOOP
33
44
COLD STAGE SENSOR
20

FIG. 2

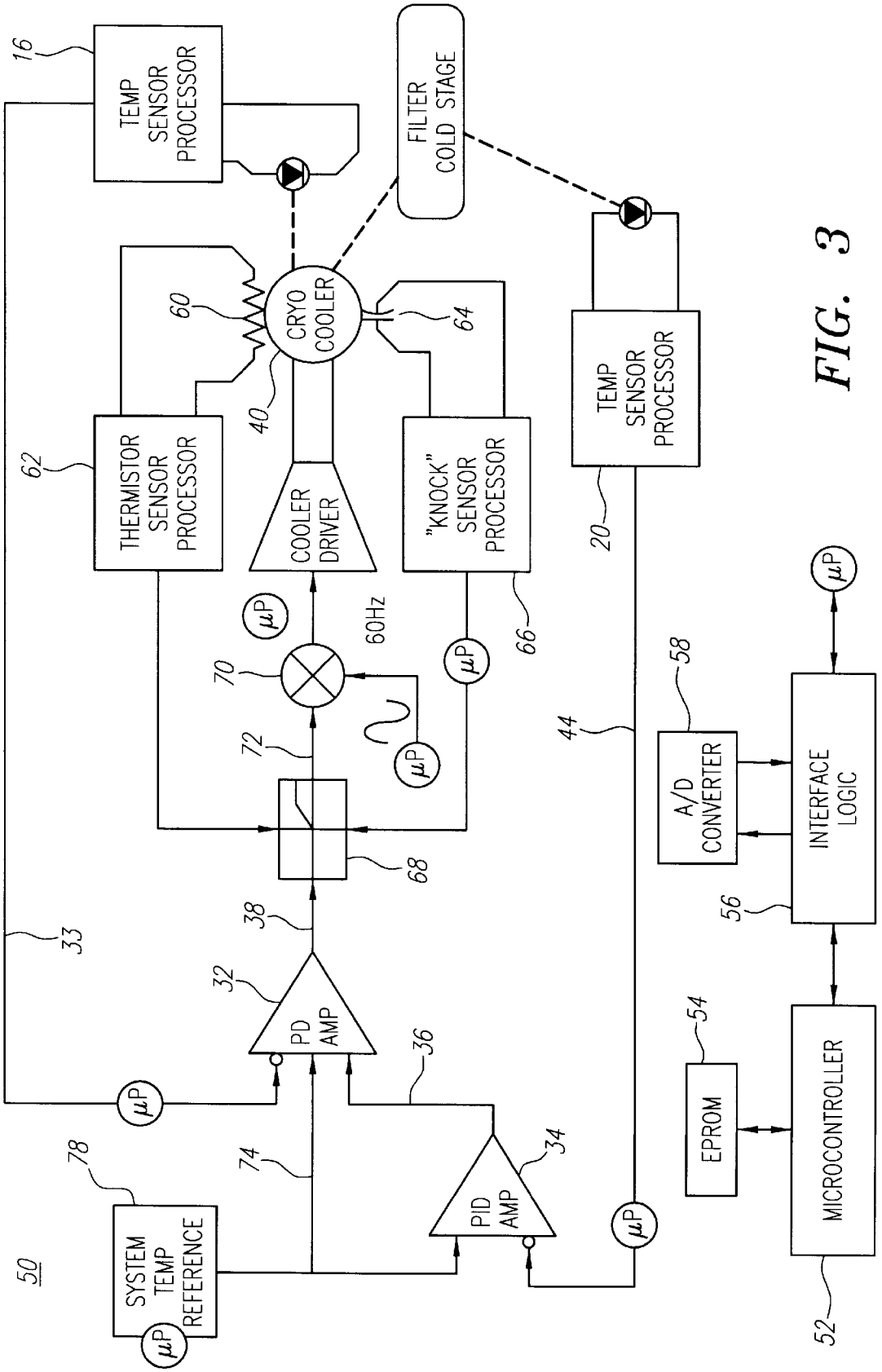

16
TEMP SENSOR PROCESSOR
FILTER COLD STAGE
60
CRYO COOLER
64
40
62
THERMISTOR SENSOR PROCESSOR
COOLER DRIVER
"KNOCK" SENSOR PROCESSOR
TEMP SENSOR PROCESSOR
20
66
70
60Hz
72
68
44
58
A/D CONVERTER
INTERFACE LOGIC
56
38
33
32
PD AMP
36
54
EPROM
MICROCONTROLLER
78
74
34
PID AMP
50
SYSTEM TEMP REFERENCE
52
μP

FIG. 3

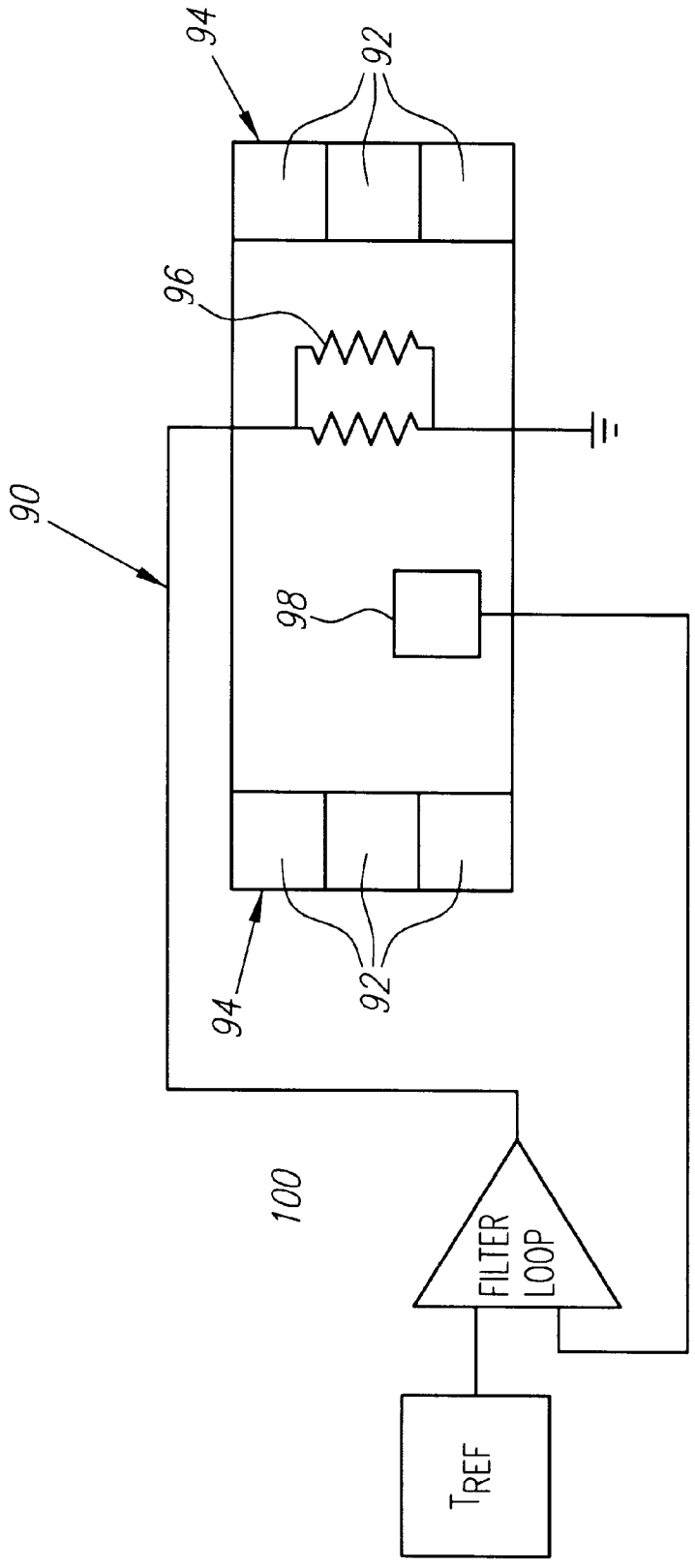
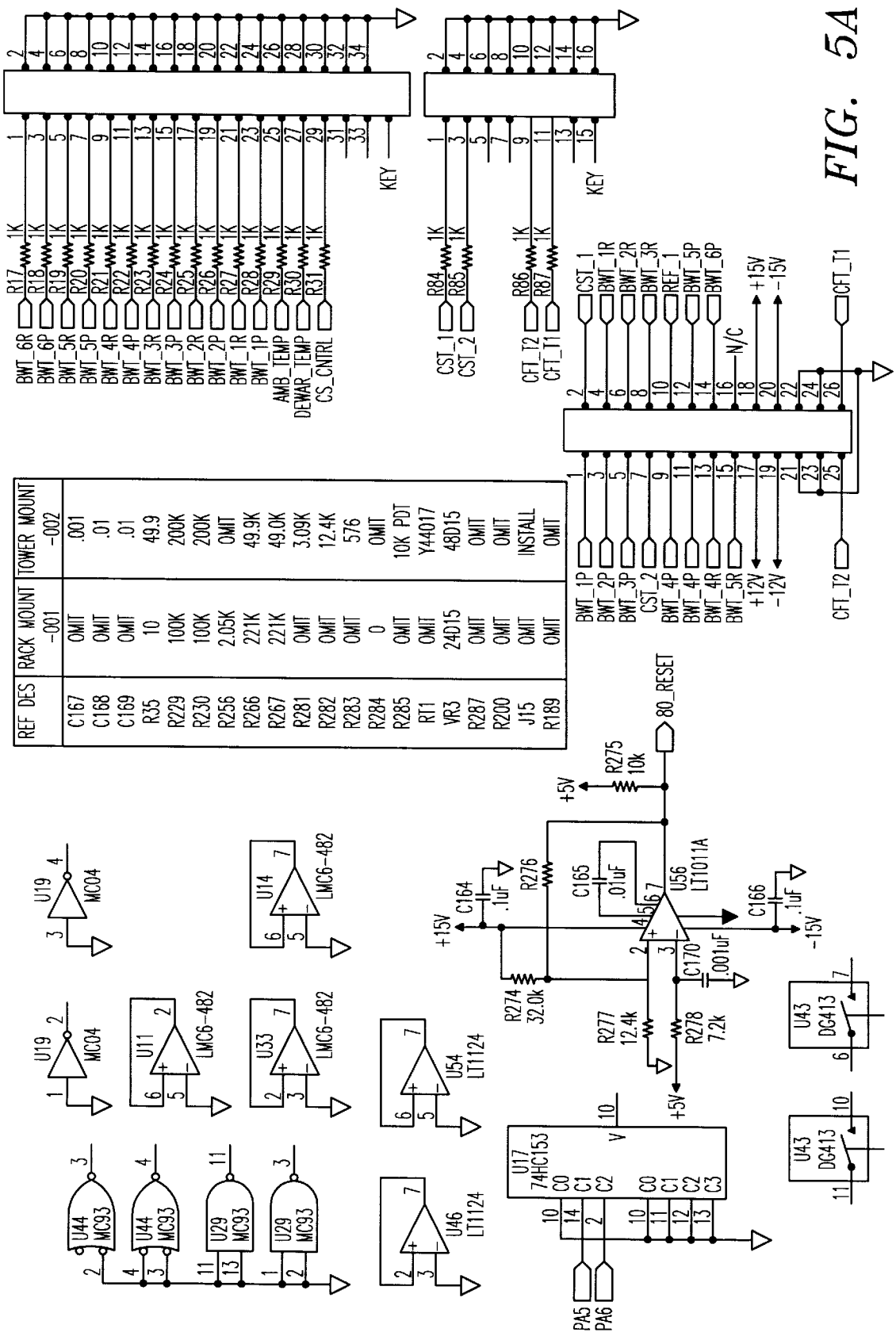


FIG. 4



REF DES	RACK MOUNT -001	TOWER MOUNT -002
C167	OMIT	.001
C168	OMIT	.01
C169	OMIT	.01
R35	10	49.9
R229	100K	200K
R230	100K	200K
R256	2.05K	OMIT
R266	221K	49.9K
R267	221K	49.0K
R281	OMIT	3.09K
R282	OMIT	12.4K
R283	OMIT	576
R284	0	OMIT
R285	OMIT	10K PDT
RT1	OMIT	Y44017
VR3	24D15	48D15
R287	OMIT	OMIT
R200	OMIT	OMIT
J15	OMIT	INSTALL
R189	OMIT	OMIT

FIG. 5A

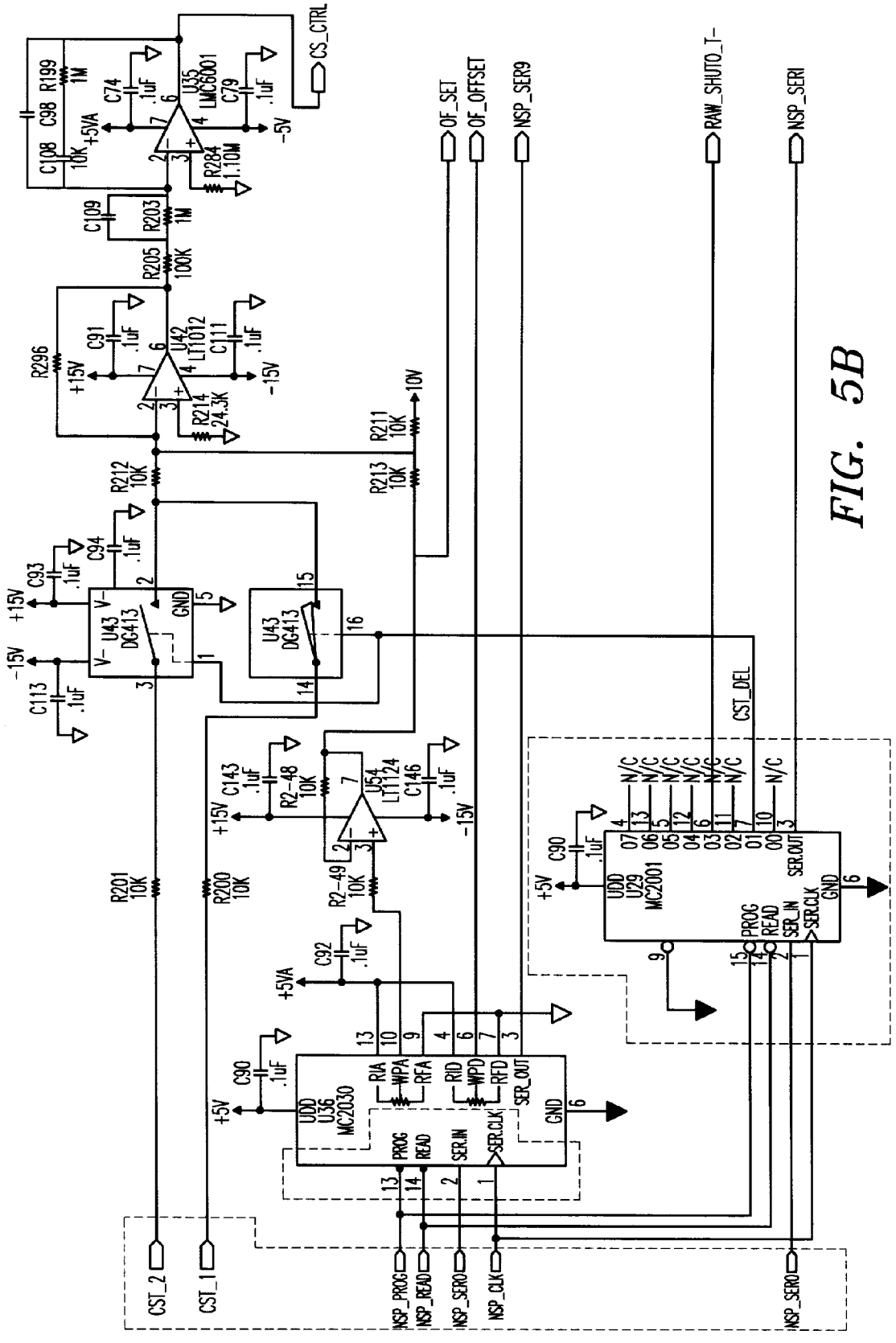

CS_CTRL
OF_SET
OF_OFFSET
NSP_SER9
RAW_SHUTO_T-
NSP_SER1
U35 LMC6001
U42 LT1012
U43 DG413
U54 LT1124
U36 MC2030
U29 MC2001
CST_DEL
CST_2
CST_1
NSP_PROG
NSP_READ
NSP_SER0
NSP_CLK
NSP_SER0

FIG. 5B

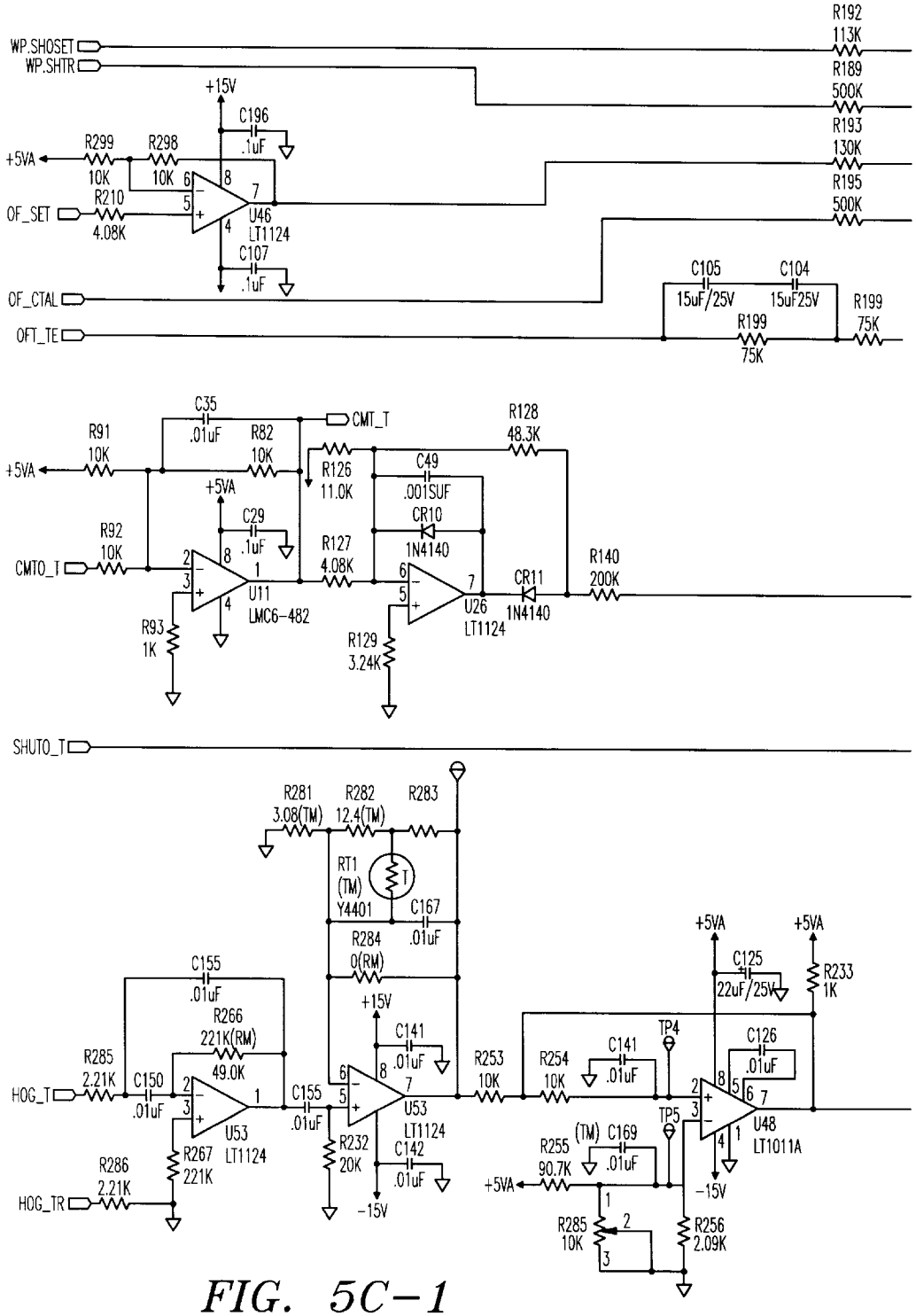

WP.SHOSET
WP.SHTR
+15V
C196
.1uF
+5VA
R299
10K
R298
10K
OF_SET
R210
4.08K
U46
LT1124
C107
.1uF
OF_CTAL
OFT_TE
R192
113K
R189
500K
R193
130K
R195
500K
C105
15uF/25V
C104
15uF25V
R199
75K
R199
75K
C35
.01uF
R91
10K
R82
10K
CMT_T
R128
48.3K
R126
11.0K
C49
.001SUF
CR10
1N4140
+5VA
C29
.1uF
R92
10K
CMTO_T
R127
4.08K
U11
LMC6-482
U26
LT1124
CR11
1N4140
R140
200K
R93
1K
R129
3.24K
SHUTO_T
R281
3.08(TM)
R282
12.4(TM)
R283
RT1
(TM)
Y4401
C167
.01uF
R284
0(RM)
+15V
C155
.01uF
R266
221K(RM)
49.0K
R285
2.21K
HOG_T
C150
.01uF
C141
.01uF
R253
10K
R254
10K
C141
.01uF
TP4
+5VA
+5VA
C125
22uF/25V
R233
1K
C126
.01uF
U53
LT1124
C155
.01uF
R232
20K
C142
.01uF
-15V
TP5
U48
LT1011A
R267
221K
R286
2.21K
HOG_TR
(TM)
C169
.01uF
R255
90.7K
+5VA
-15V
R285
10K
R256
2.09K

FIG. 5C-1

FIG. 5C-2

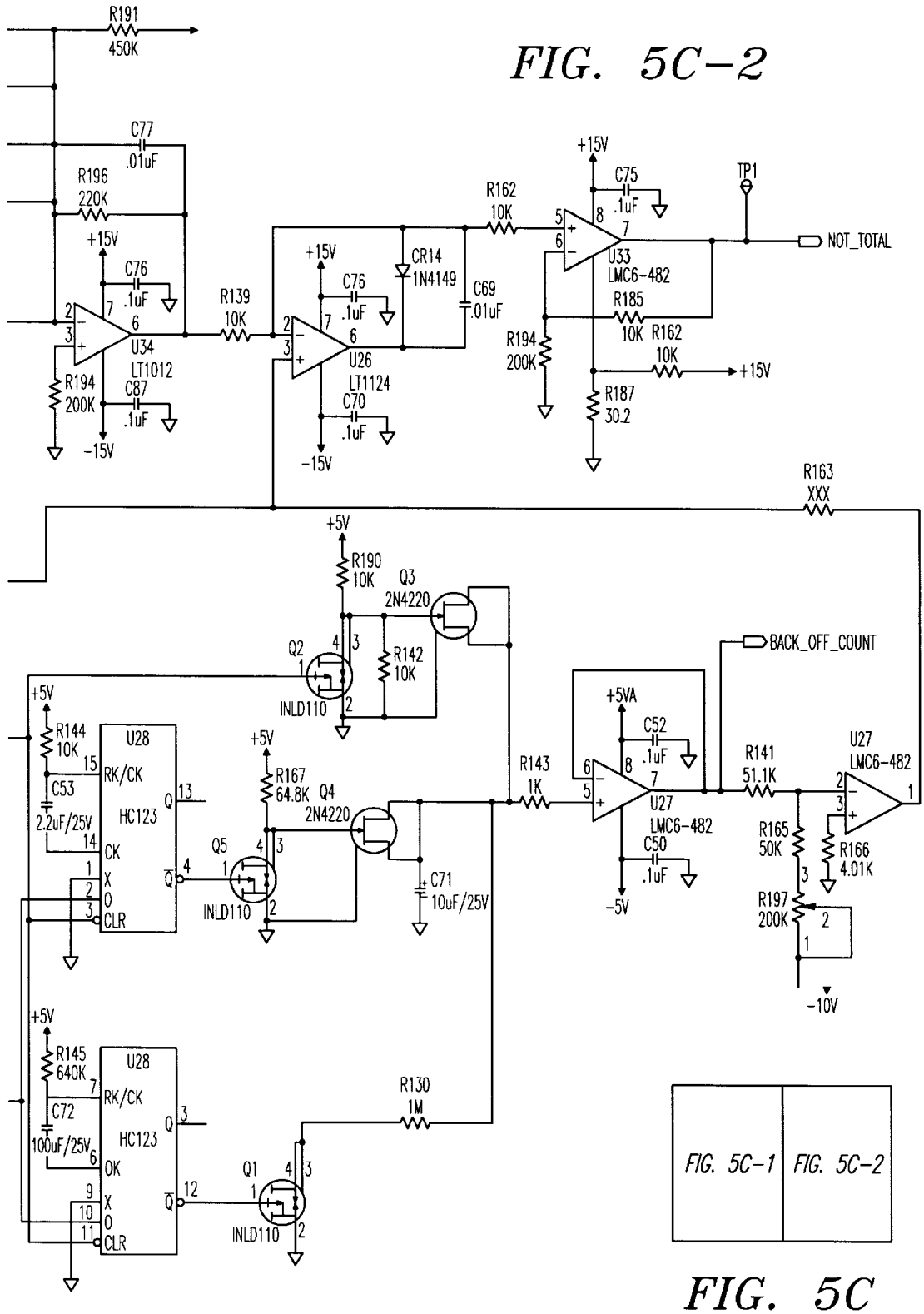


FIG. 5C-1	FIG. 5C-2

FIG. 5C

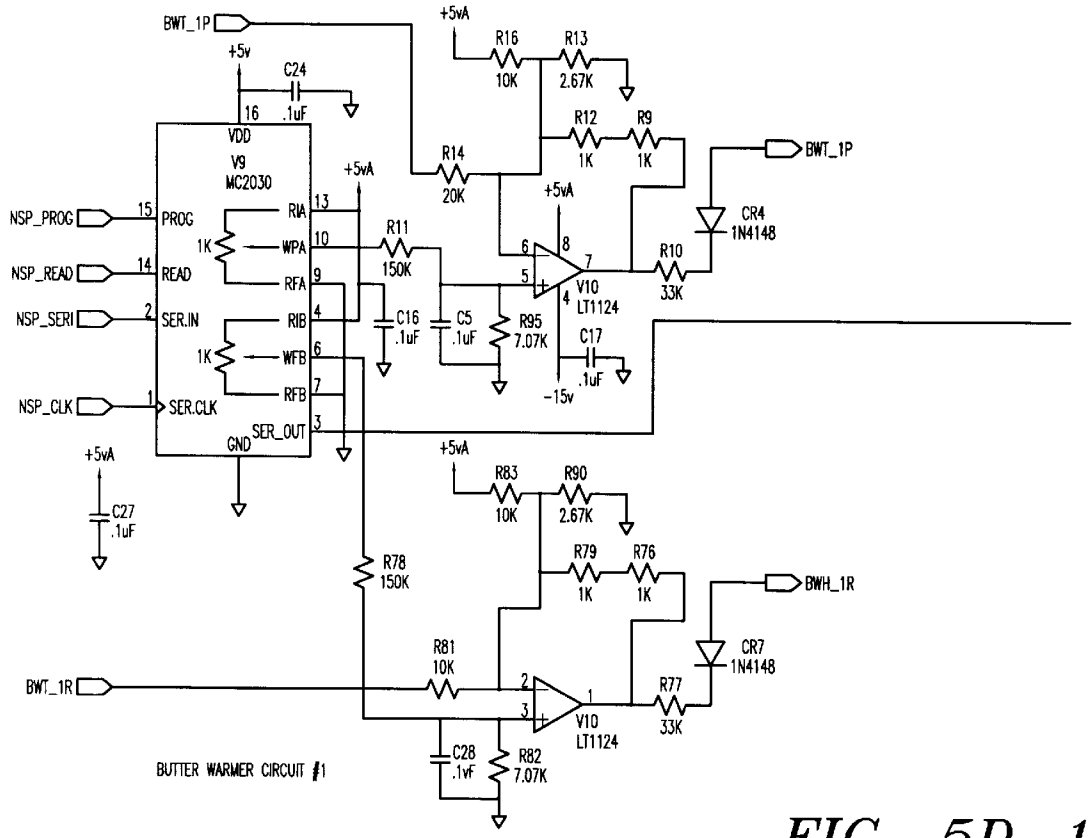


FIG. 5D-1

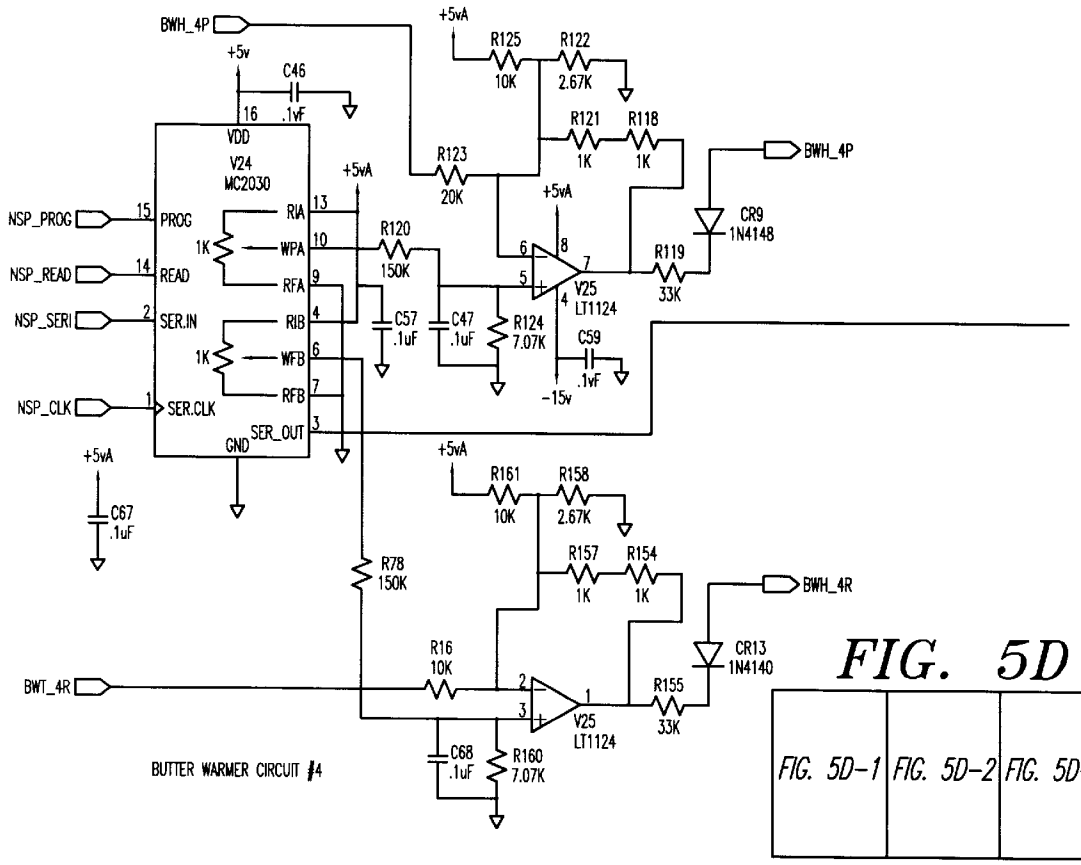


FIG. 5D

FIG. 5D-1	FIG. 5D-2	FIG. 5D-3

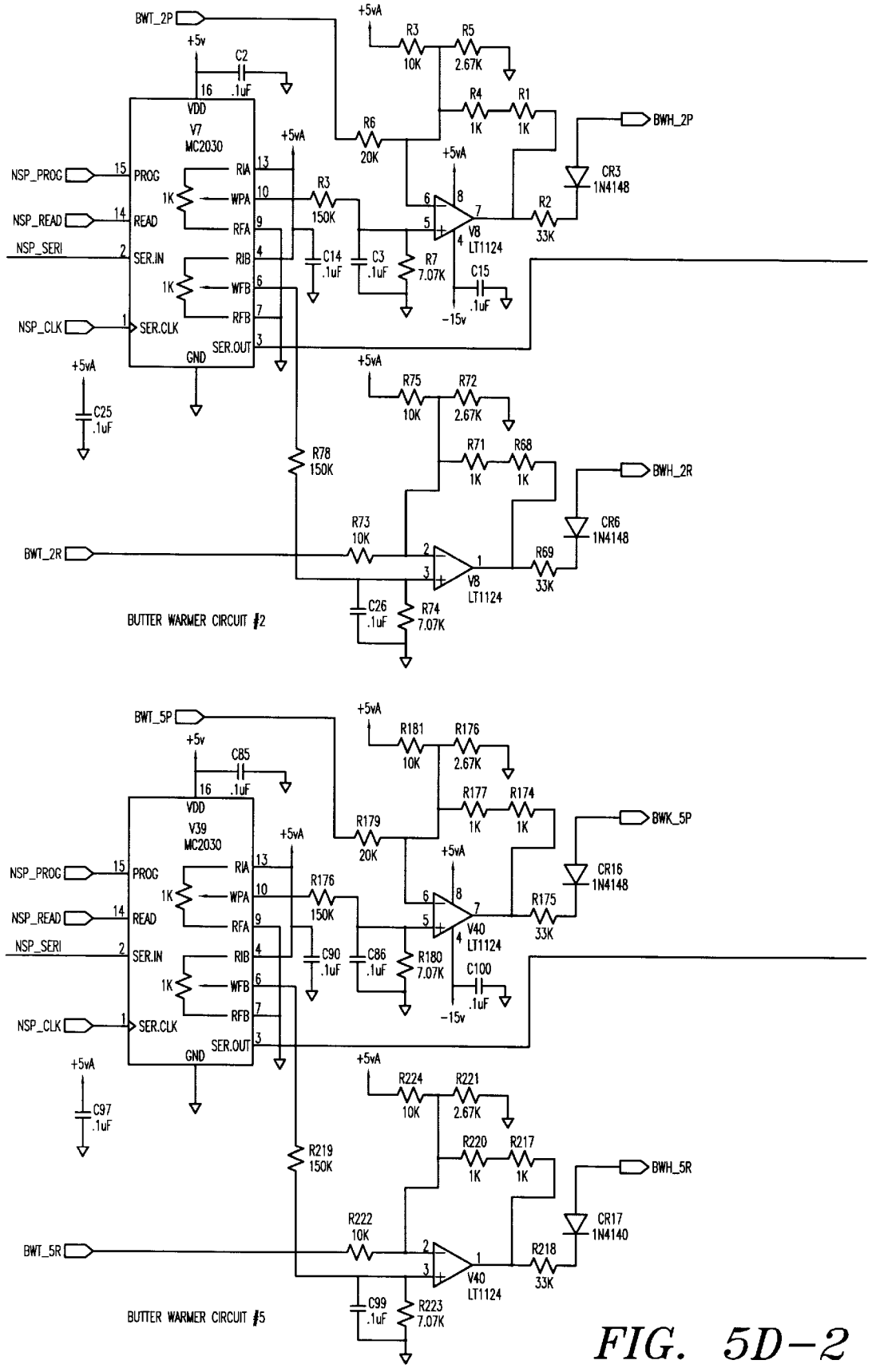

BWT_2P
+5v
C2
.1uF
16
VDD
V7
MC2030
+5vA
R3
10K
R5
2.67K
R4
1K
R1
1K
R6
20K
BWH_2P
CR3
1N4148
NSP_PROG
15
PROG
RIA
13
1K
WPA
10
R3
150K
NSP_READ
14
READ
RFA
9
R2
33K
NSP_SERI
2
SER.IN
RIB
4
C14
.1uF
C3
.1uF
R7
7.07K
C15
.1uF
-15v
V8
LT1124
1K
WFB
6
NSP_CLK
1
SER.CLK
RFB
7
SER.OUT
3
GND
C25
.1uF
R75
10K
R72
2.67K
R78
150K
R71
1K
R68
1K
BWH_2R
CR6
1N4148
BWT_2R
R73
10K
R69
33K
C26
.1uF
R74
7.07K
BUTTER WARMER CIRCUIT #2
BWT_5P
C85
R181
R176
R177
R174
BWK_5P
R179
CR16
1N4148
R176
150K
R175
C90
C86
R180
C100
V40
LT1124
C97
R219
150K
R224
R221
R220
R217
BWH_5R
CR17
1N4140
R222
R218
C99
R223
BUTTER WARMER CIRCUIT #5

FIG. 5D-2

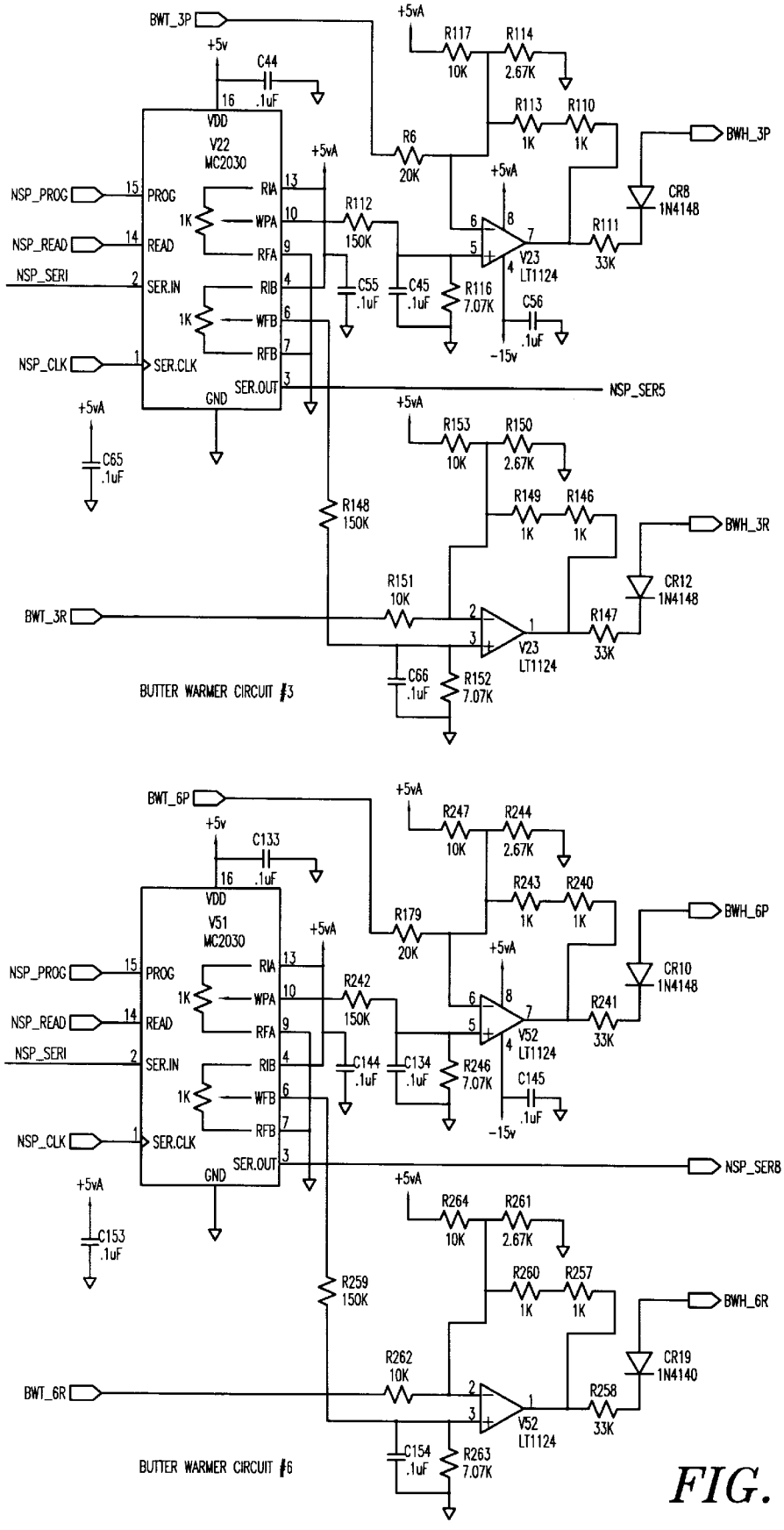

BWT_3P
+5v
C44
.1uF
VDD
V22
MC2030
+5vA
R6
20K
R117
10K
R114
2.67K
R113
1K
R110
1K
BWH_3P
CR8
1N4148
NSP_PROG
PROG
RIA
WPA
RFA
RIB
WFB
RFB
NSP_READ
READ
NSP_SERI
SER.IN
NSP_CLK
SER.CLK
SER.OUT
GND
R112
150K
C55
.1uF
C45
.1uF
R116
7.07K
C56
.1uF
-15v
V23
LT1124
R111
33K
NSP_SER5
C65
.1uF
R148
150K
R153
10K
R150
2.67K
R149
1K
R146
1K
BWH_3R
CR12
1N4148
BWT_3R
R151
10K
R147
33K
C66
.1uF
R152
7.07K
BUTTER WARMER CIRCUIT #3
BWT_6P
C133
R247
10K
R244
2.67K
R243
1K
R240
1K
BWH_6P
CR10
1N4148
R179
20K
R242
150K
R241
33K
C144
.1uF
C134
.1uF
R246
7.07K
C145
.1uF
V52
LT1124
NSP_SER8
C153
.1uF
R259
150K
R264
10K
R261
2.67K
R260
1K
R257
1K
BWH_6R
CR19
1N4140
R262
10K
R258
33K
C154
.1uF
R263
7.07K
BUTTER WARMER CIRCUIT #6

FIG. 5D-3

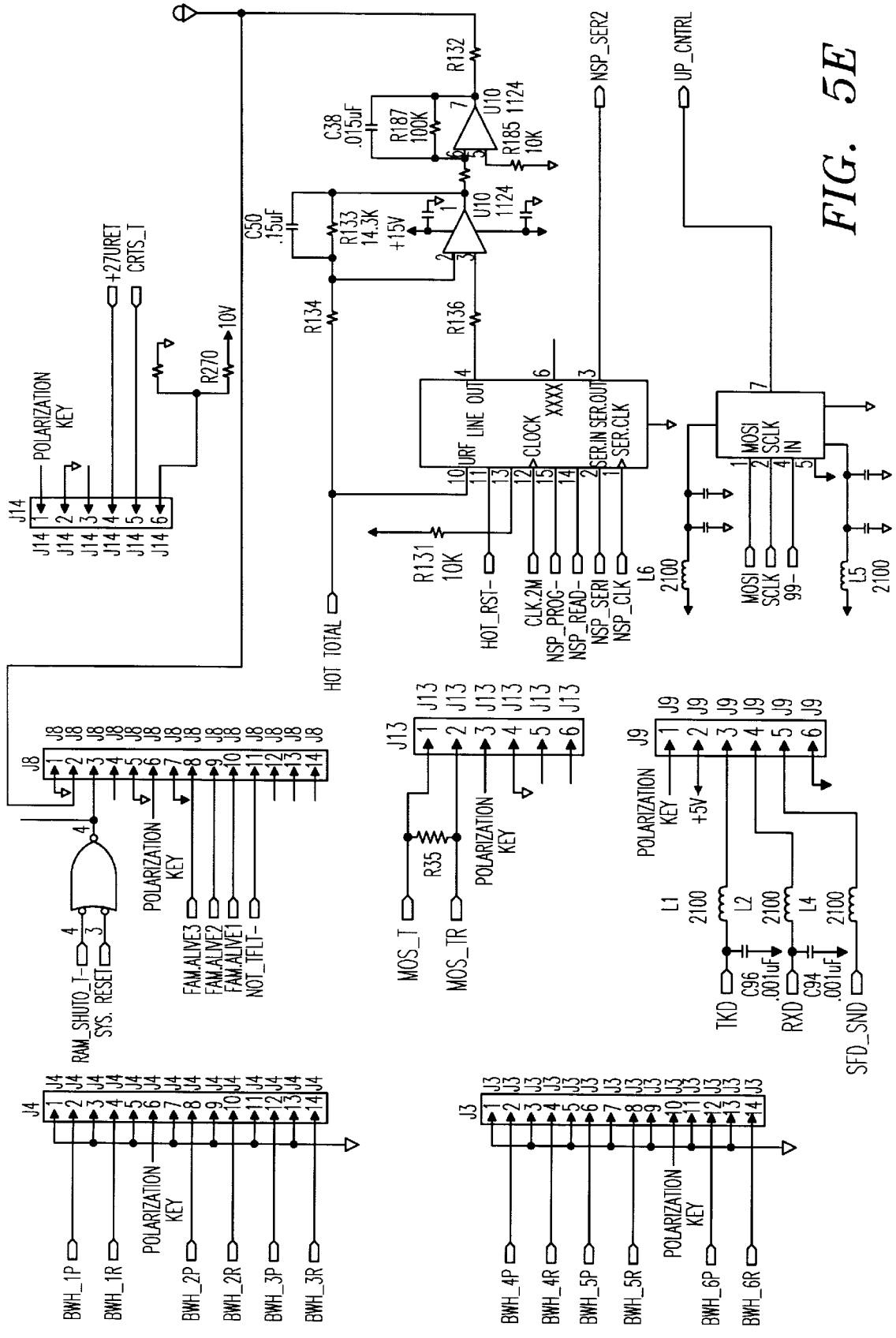
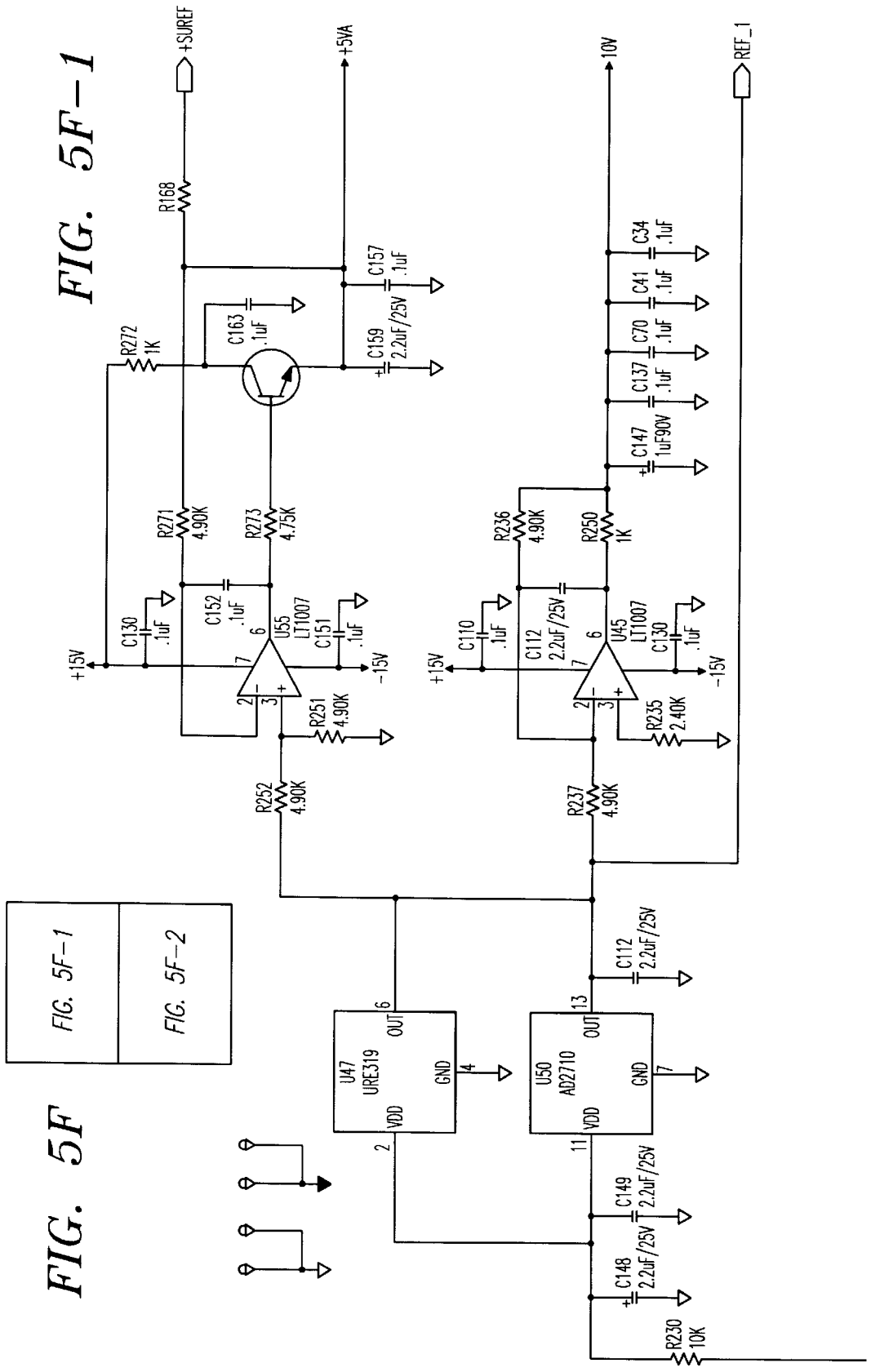

J14
POLARIZATION KEY
+27URET
CRTS_T
R270
10V
R134
R136
R133 14.3K
C50 .15uF
+15V
U10 1124
R132
C38 .015uF
R187 100K
R185 10K
NSP_SER2
UP_CNTRL
URF LINE OUT
CLOCK
XXXX
SER.IN SER.OUT
SER.CLK
MOSI
SCLK
IN
R131 10K
HOT TOTAL
HOT_RST-
CLK.2M
NSP_PROG-
NSP_READ-
NSP_SERI
NSP_CLK
L6 2100
L5 2100
J8
RAM_SHUTO_T-
SYS. RESET
FAM.ALIVE3
FAM.ALIVE2
FAM.ALIVE1
NOT_TFLT-
J13
R35
MOS_T
MOS_TR
J9
+5V
L1 2100
L2 2100
L4 2100
C96 .001uF
C94 .001uF
TKD
RXD
SFD_SND
J4
BWH_1P
BWH_1R
BWH_2P
BWH_2R
BWH_3P
BWH_3R
J3
BWH_4P
BWH_4R
BWH_5P
BWH_5R
BWH_6P
BWH_6R

FIG. 5E

FIG. 5F


FIG. 5F-1
FIG. 5F-2
+SUREF
+5VA
10V
REF_1
R168
R272 1K
C163 .1uF
C157 .1uF
C159 2.2uF/25V
R271 4.90K
R273 4.75K
C130 .1uF
C152 .1uF
C151 .1uF
U55 LT1007
+15V
-15V
R251 4.90K
R252 4.90K
R236 4.90K
R250 1K
C110 .1uF
C112 2.2uF/25V
C130 .1uF
U45 LT1007
R235 2.40K
R237 4.90K
C34 .1uF
C41 .1uF
C70 .1uF
C137 .1uF
C147 1uF90V
U47 URE319
U50 AD2710
VDD
OUT
GND
C112 2.2uF/25V
C149 2.2uF/25V
C148 2.2uF/25V
R230 10K

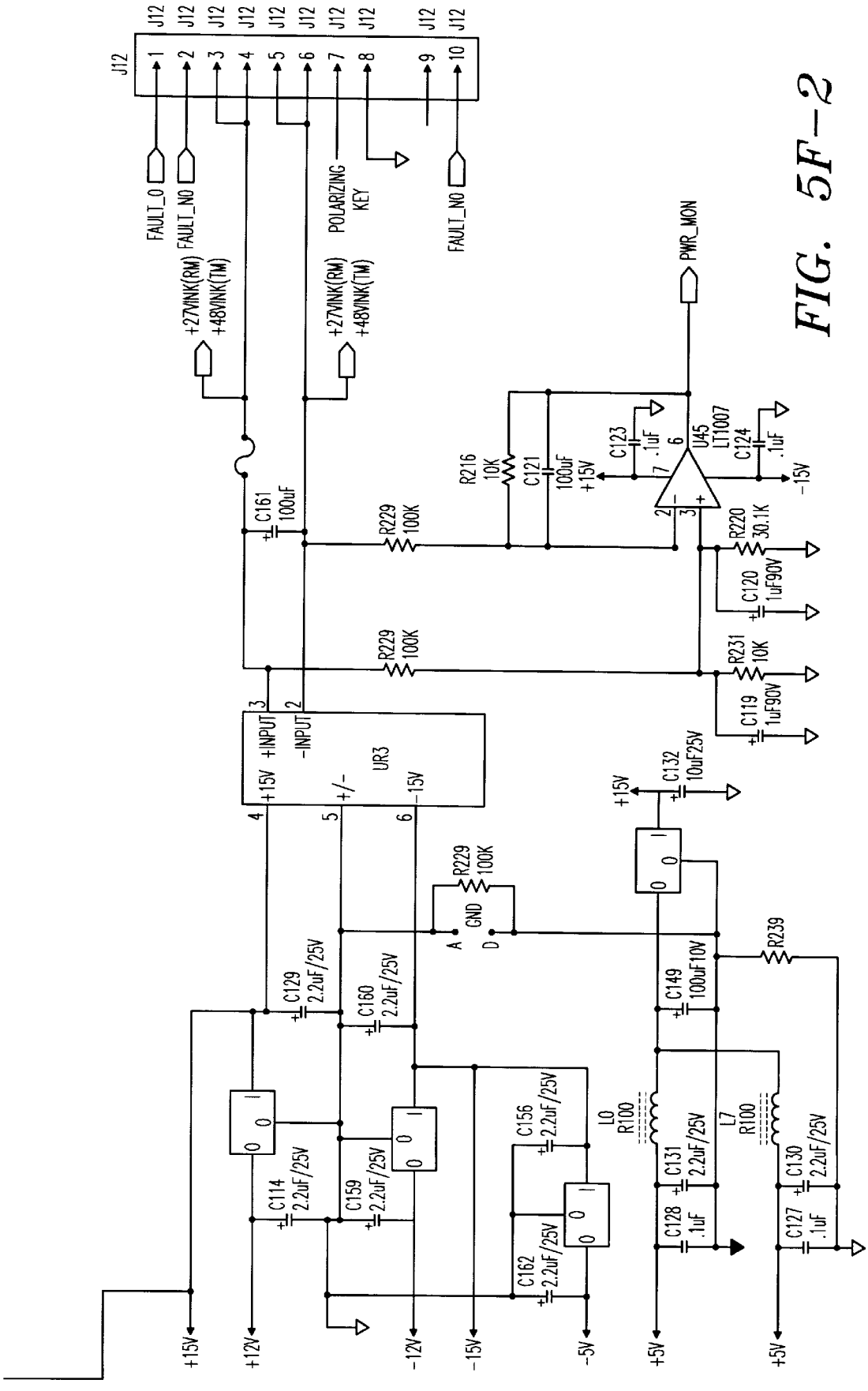

J12
FAULT_0
FAULT_N0
+27VINK(RM)
+48VINK(TM)
POLARIZING KEY
FAULT_N0
PWR_MON
R216 10K
C121 100uF
C123 .1uF
U45 LT1007
C124 .1uF
C161 100uF
R229 100K
R220 30.1K
C120 1uF90V
R231 10K
C119 1uF90V
UR3
+INPUT
-INPUT
+15V
+/-
-15V
C132 10uF25V
GND
C129 2.2uF/25V
C160 2.2uF/25V
C114 2.2uF/25V
C159 2.2uF/25V
C156 2.2uF/25V
C162 2.2uF/25V
C149 100uF10V
R239
L0 R100
L7 R100
C131 2.2uF/25V
C130 2.2uF/25V
C128 .1uF
C127 .1uF
+15V
+12V
-12V
-15V
-5V
+5V

FIG. 5F-2

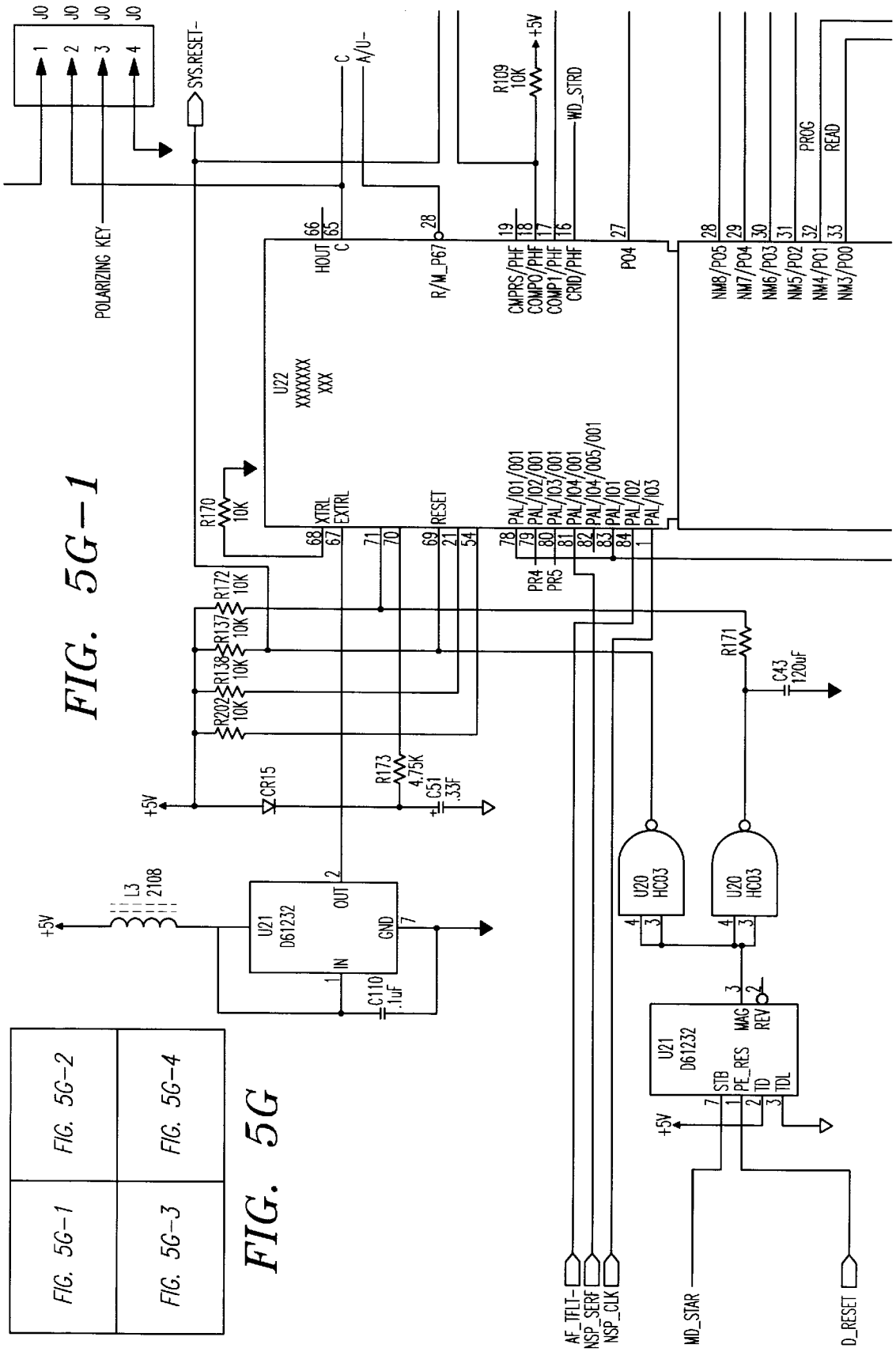


FIG. 5G-1

FIG. 5G-1	FIG. 5G-2
FIG. 5G-3	FIG. 5G-4

FIG. 5G

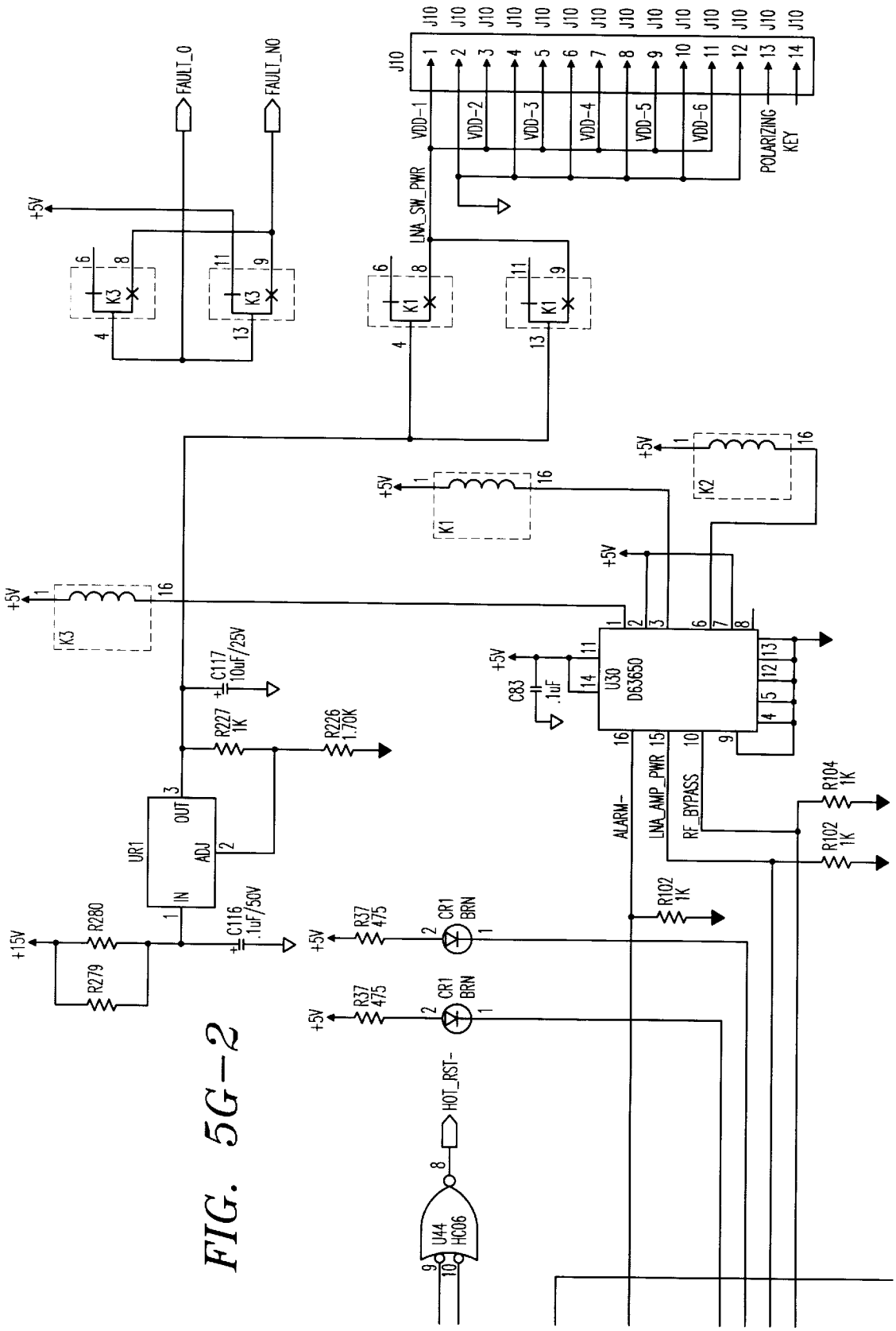

J10
VDD-1
VDD-2
VDD-3
VDD-4
VDD-5
VDD-6
POLARIZING KEY
FAULT_O
FAULT_NO
LNA_SW_PWR
+5V
+15V
K1
K2
K3
U30 D63650
C83 .1uF
C117 10uF/25V
C116 .1uF/50V
R227 1K
R226 1.70K
R280
R279
R37 475
CR1 BRN
UR1
OUT
ADJ
IN
ALARM-
LNA_AMP_PWR
RF_BYPASS
R104 1K
R102 1K
HOT_RST-
U44 HC06

FIG. 5G-2

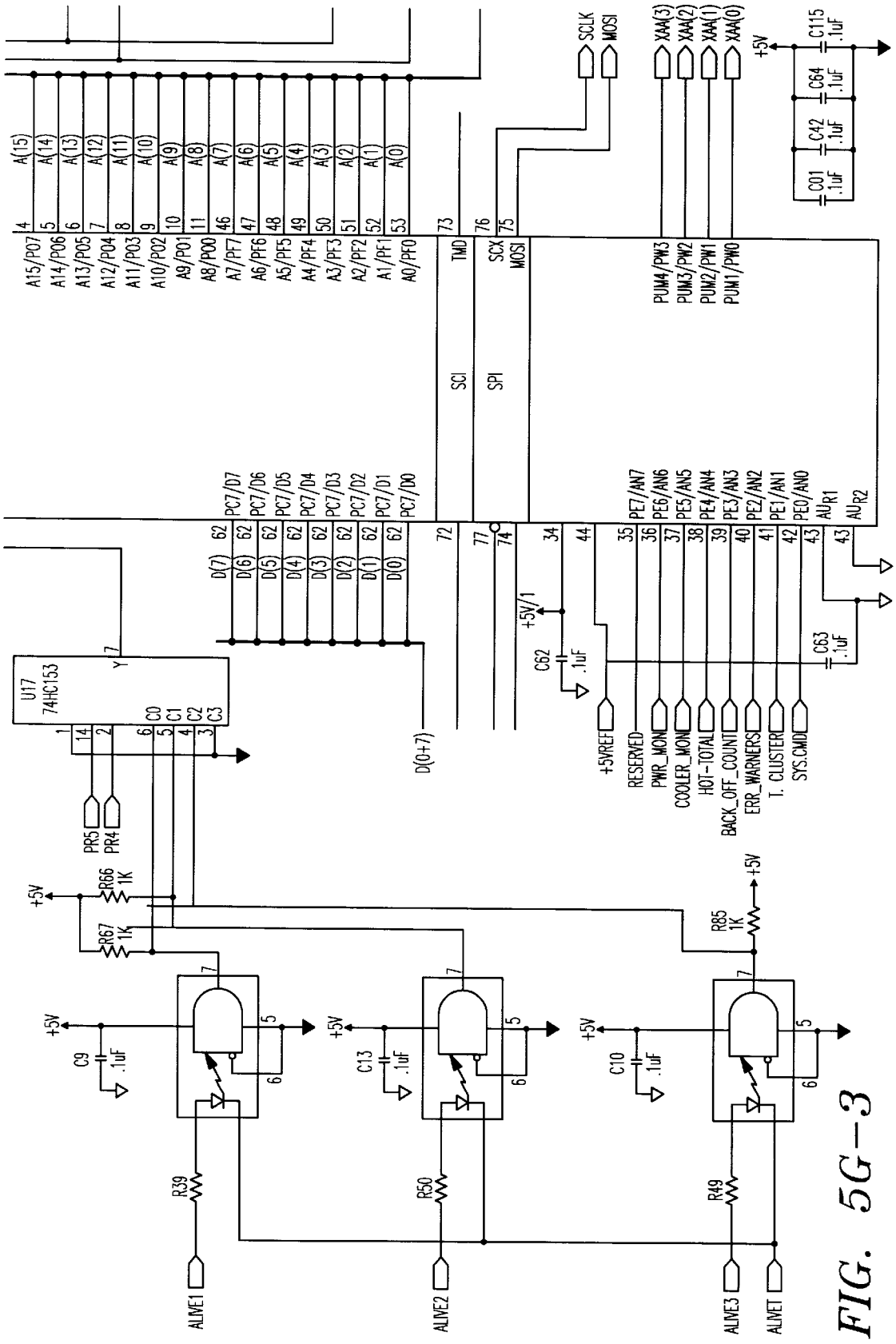

A(15)
A(14)
A(13)
A(12)
A(11)
A(10)
A(9)
A(8)
A(7)
A(6)
A(5)
A(4)
A(3)
A(2)
A(1)
A(0)
A15/PD7
A14/PD6
A13/PD5
A12/PD4
A11/PD3
A10/PD2
A9/PD1
A8/PD0
A7/PF7
A6/PF6
A5/PF5
A4/PF4
A3/PF3
A2/PF2
A1/PF1
A0/PF0
TMD
SCX
MOSI
SCLK
MOSI
SCI
SPI
XAA(3)
XAA(2)
XAA(1)
XAA(0)
PUM4/PW3
PUM3/PW2
PUM2/PW1
PUM1/PW0
+5V
C115 .1uF
C64 .1uF
C42 .1uF
C01 .1uF
PC7/D7
PC7/D6
PC7/D5
PC7/D4
PC7/D3
PC7/D2
PC7/D1
PC7/D0
D(7)
D(6)
D(5)
D(4)
D(3)
D(2)
D(1)
D(0)
D(0+7)
+5V/1
C62 .1uF
PE7/AN7
PE6/AN6
PE5/AN5
PE4/AN4
PE3/AN3
PE2/AN2
PE1/AN1
PE0/AN0
AUR1
AUR2
C63 .1uF
+5VREF
RESERVED
PWR_MON
COOLER_MON
HOT-TOTAL
BACK_OFF_COUNT
ERR_WARNERS
T. CLUSTER
SYS.CMD
U17
74HC153
C0
C1
C2
C3
PR5
PR4
R66 1K
R67 1K
R85 1K
C9 .1uF
C13 .1uF
C10 .1uF
R39
R50
R49
ALME1
ALME2
ALME3
ALMET

FIG. 5G-3

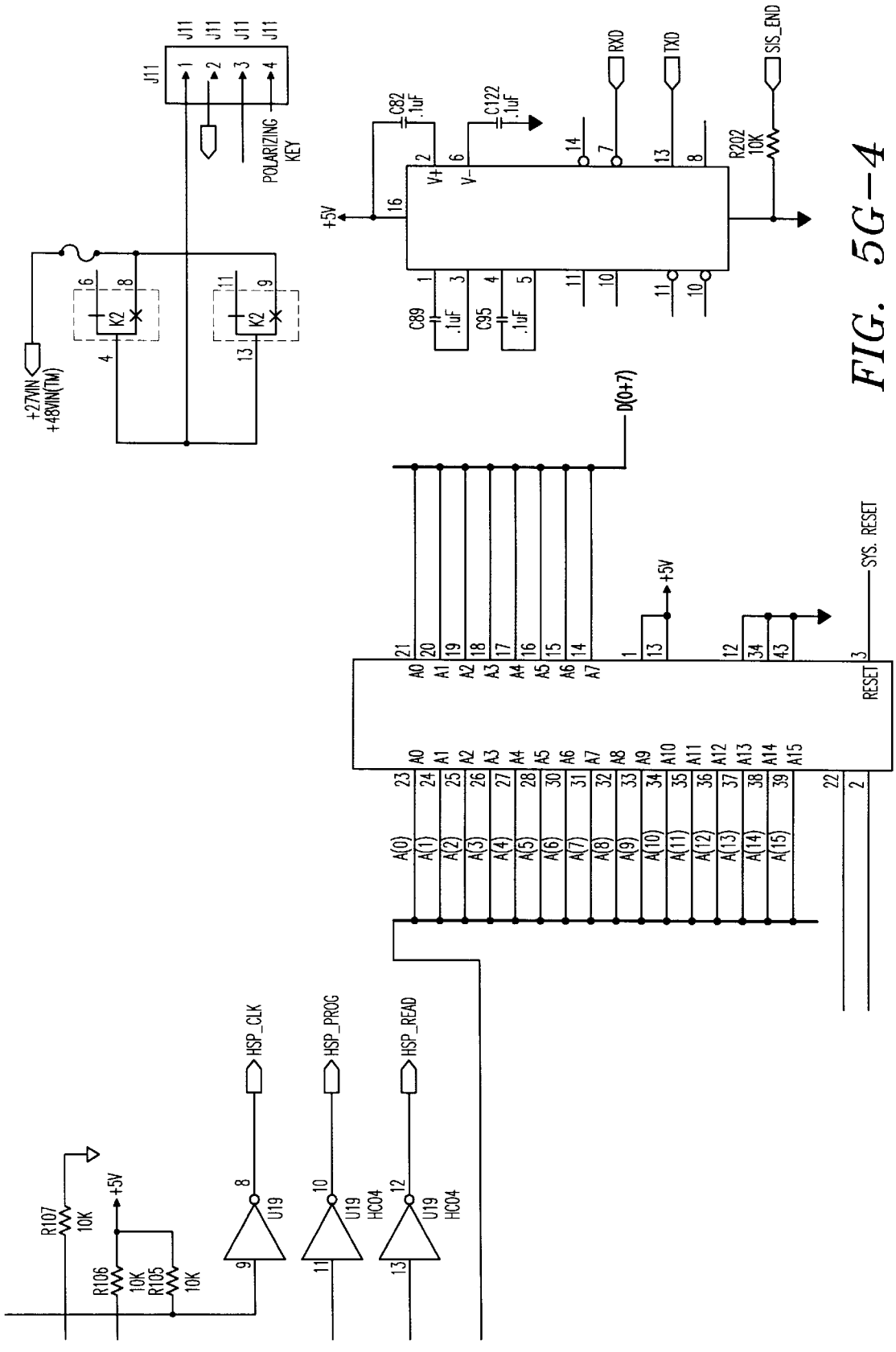

J11
POLARIZING KEY
+27VIN
+48VIN(TM)
K2
C82 .1uF
C122 .1uF
C89 .1uF
C95 .1uF
+5V
V+
V-
RXD
TXD
SIS_END
R202 10K
D(0+7)
A0
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
A13
A14
A15
RESET
SYS. RESET
R107 10K
R106 10K
R105 10K
HSP_CLK
HSP_PROC
HSP_READ
U19
HC04

FIG. 5G-4

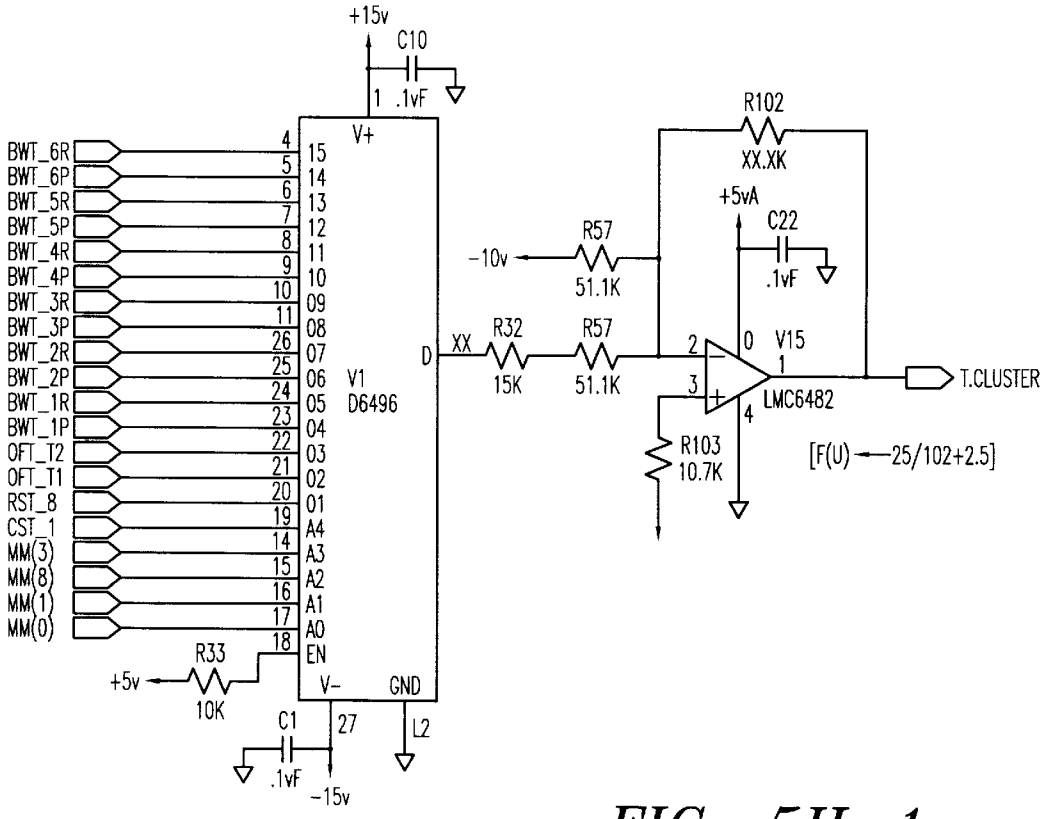


FIG. 5H-1

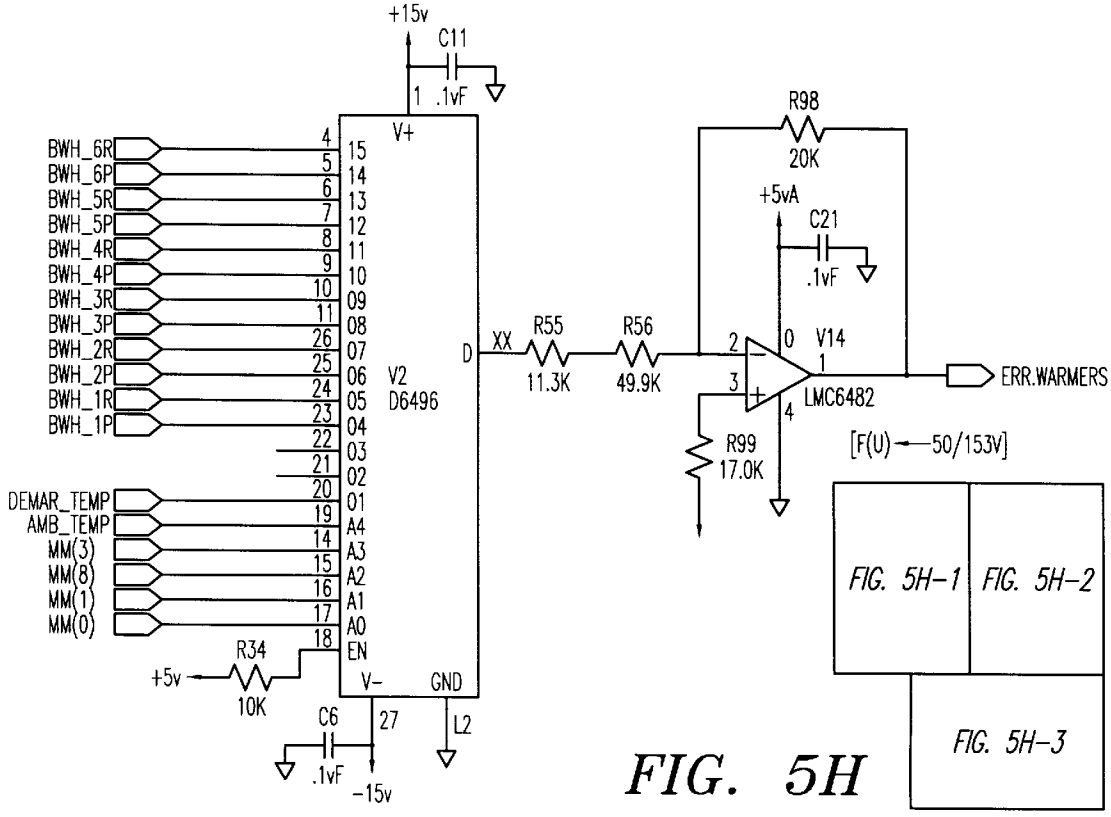


FIG. 5H

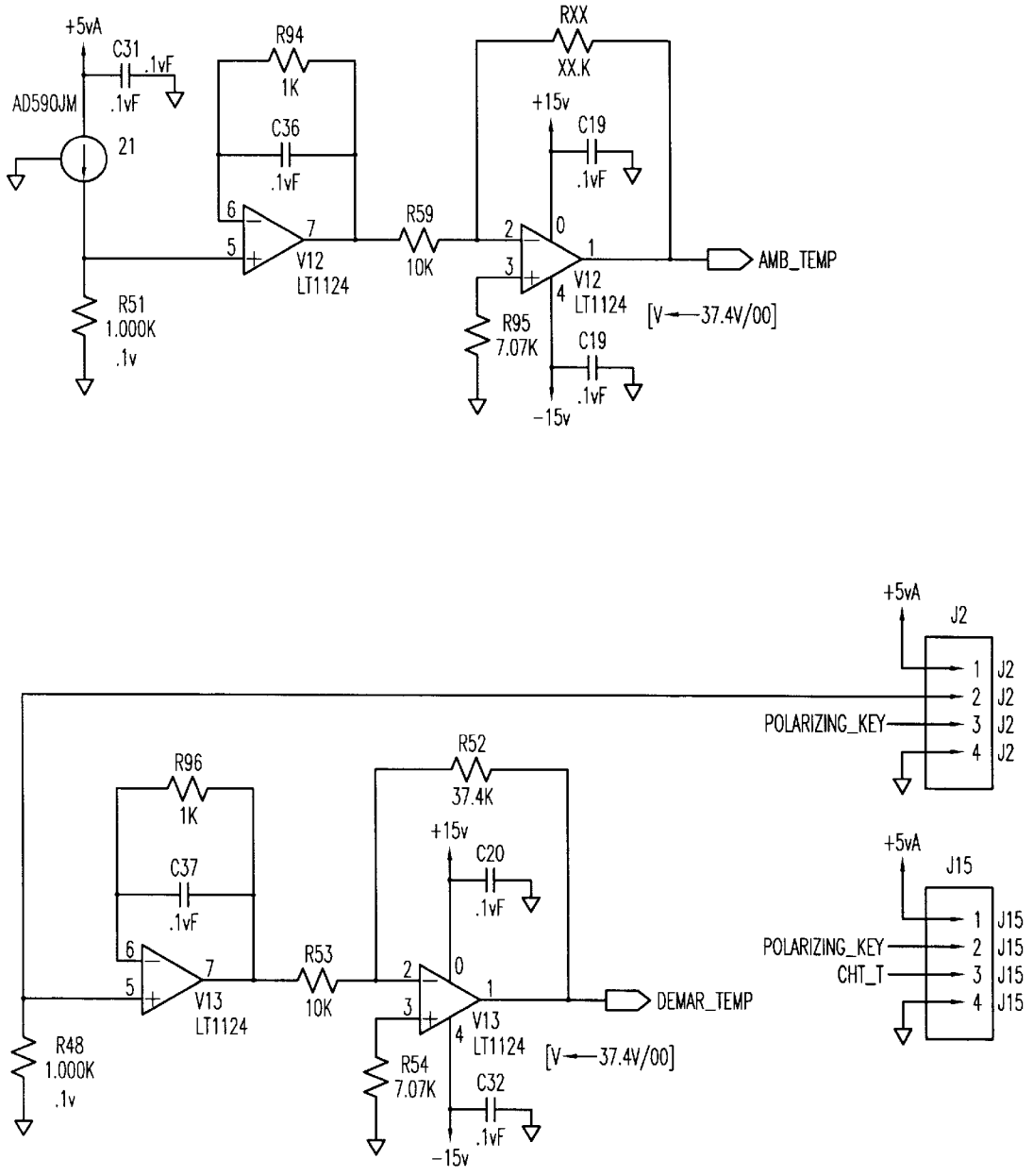

+5vA
C31
.1vF
AD590JM
.1vF
21
R94
1K
C36
.1vF
RXX
XX.K
+15v
C19
.1vF
6
5
7
V12
LT1124
R59
10K
2
3
0
1
4
V12
LT1124
AMB_TEMP
[V←37.4V/00]
R51
1.000K
.1v
R95
7.07K
C19
.1vF
-15v
+5vA
J2
1 J2
2 J2
POLARIZING_KEY
3 J2
4 J2
R52
37.4K
R96
1K
+15v
C20
.1vF
C37
.1vF
+5vA
J15
1 J15
POLARIZING_KEY
2 J15
CHT_T
3 J15
4 J15
6
5
7
V13
LT1124
R53
10K
2
3
0
1
4
V13
LT1124
DEMAR_TEMP
[V←37.4V/00]
R48
1.000K
.1v
R54
7.07K
C32
.1vF
-15v

FIG. 5H-2

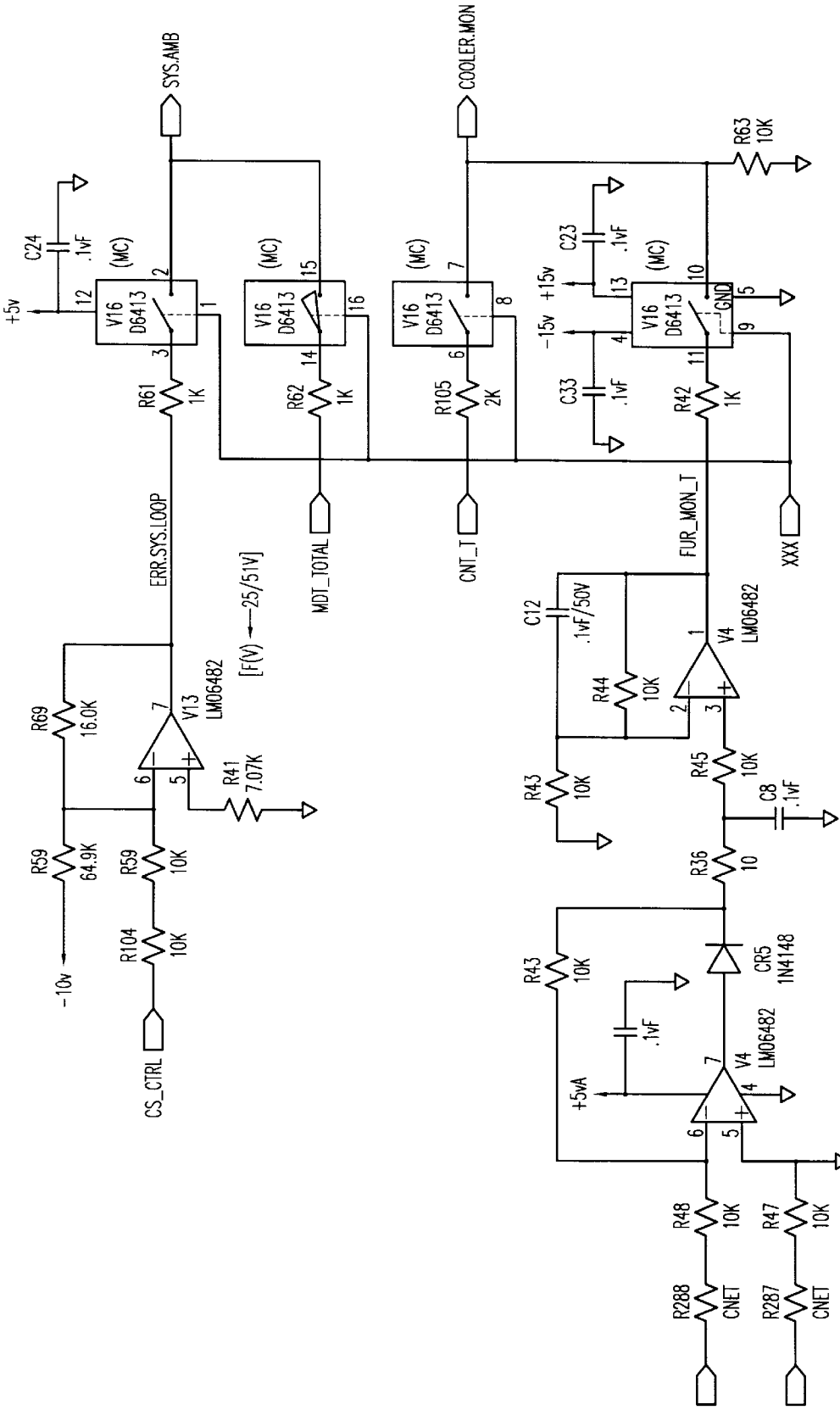

SYS.AMB
COOLER.MON
+5v
C24
.1vF
V16
D6413
(MC)
R61
1K
R62
1K
R105
2K
C23
.1vF
+15v
-15v
C33
.1vF
R42
1K
R63
10K
GND
ERR.SYS.LOOP
MDI_TOTAL
CNT_T
FUR_MON_T
XXX
[F(V) → 25/51V]
V13
LM06482
R69
16.0K
R41
7.07K
R59
64.9K
R59
10K
R104
10K
-10v
CS_CTRL
C12
.1vF/50V
R44
10K
V4
LM06482
R43
10K
R45
10K
C8
.1vF
R36
10
CR5
1N4148
+5vA
R48
10K
R47
10K
R288
CNET
R287
CNET

FIG. 5H-3

FIG. 6A

FIG. 6A-1	FIG. 6A-2

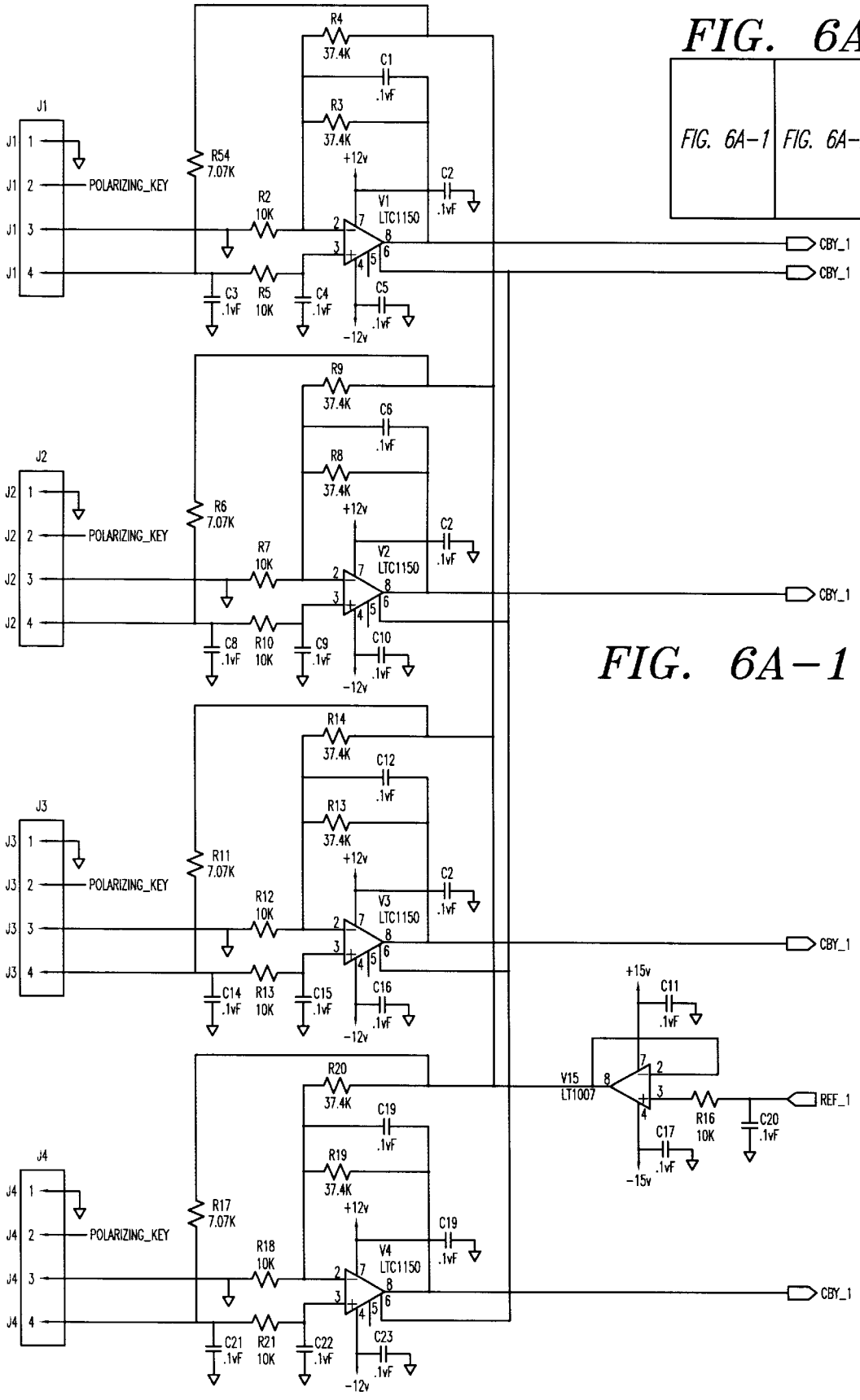


FIG. 6A-1

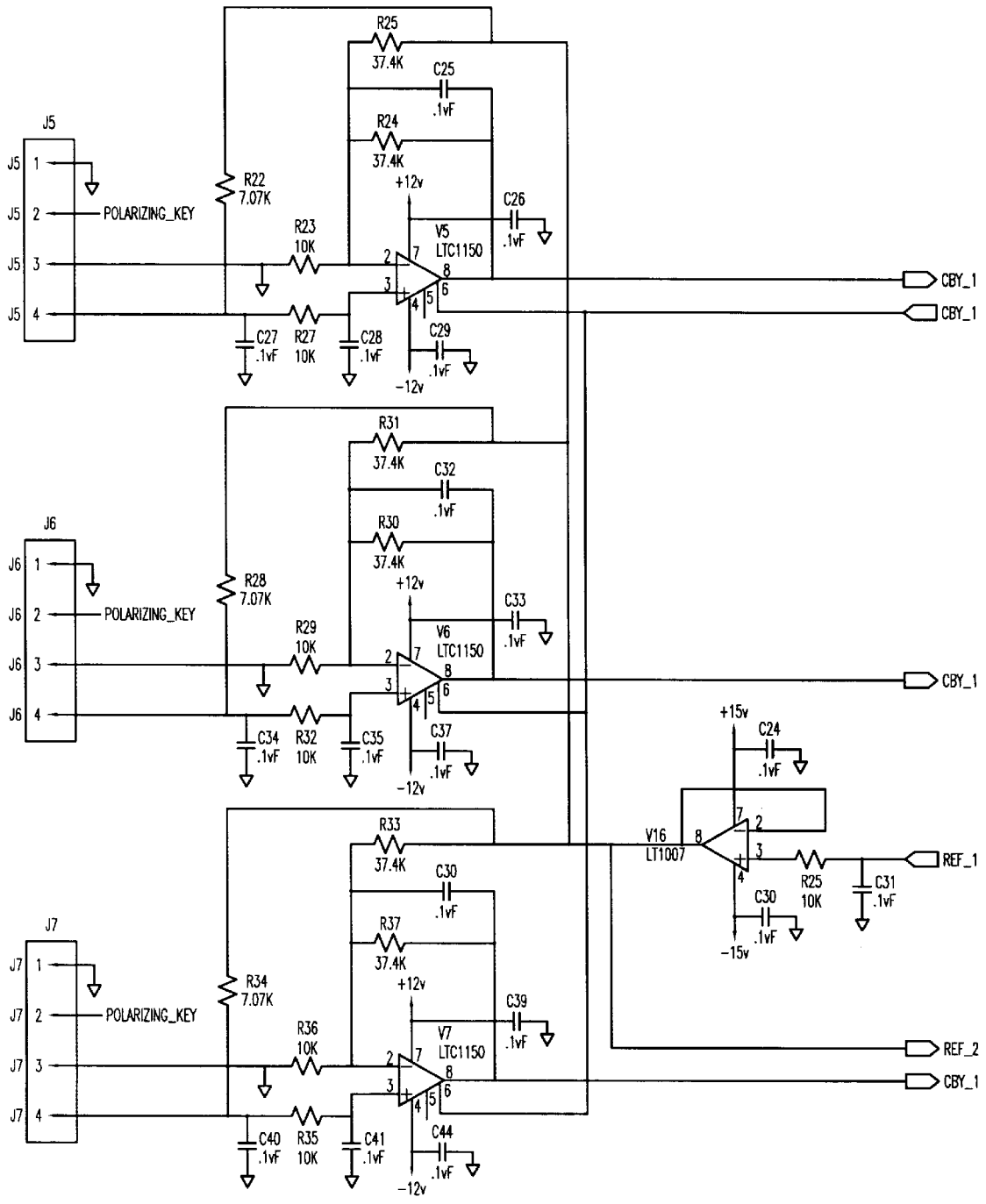

J5
J5 1
J5 2
J5 3
J5 4
POLARIZING_KEY
R22 7.07K
R23 10K
R24 37.4K
R25 37.4K
C25 .1vF
C26 .1vF
V5 LTC1150
+12v
-12v
C27 .1vF
R27 10K
C28 .1vF
C29 .1vF
CBY_1
CBY_1
J6
J6 1
J6 2
J6 3
J6 4
POLARIZING_KEY
R28 7.07K
R29 10K
R30 37.4K
R31 37.4K
C32 .1vF
C33 .1vF
V6 LTC1150
+12v
-12v
C34 .1vF
R32 10K
C35 .1vF
C37 .1vF
CBY_1
J7
J7 1
J7 2
J7 3
J7 4
POLARIZING_KEY
R34 7.07K
R36 10K
R37 37.4K
R33 37.4K
C30 .1vF
C39 .1vF
V7 LTC1150
+12v
-12v
C40 .1vF
R35 10K
C41 .1vF
C44 .1vF
CBY_1
REF_2
+15v
V16 LT1007
C24 .1vF
C30 .1vF
-15v
R25 10K
C31 .1vF
REF_1

FIG. 6A–2

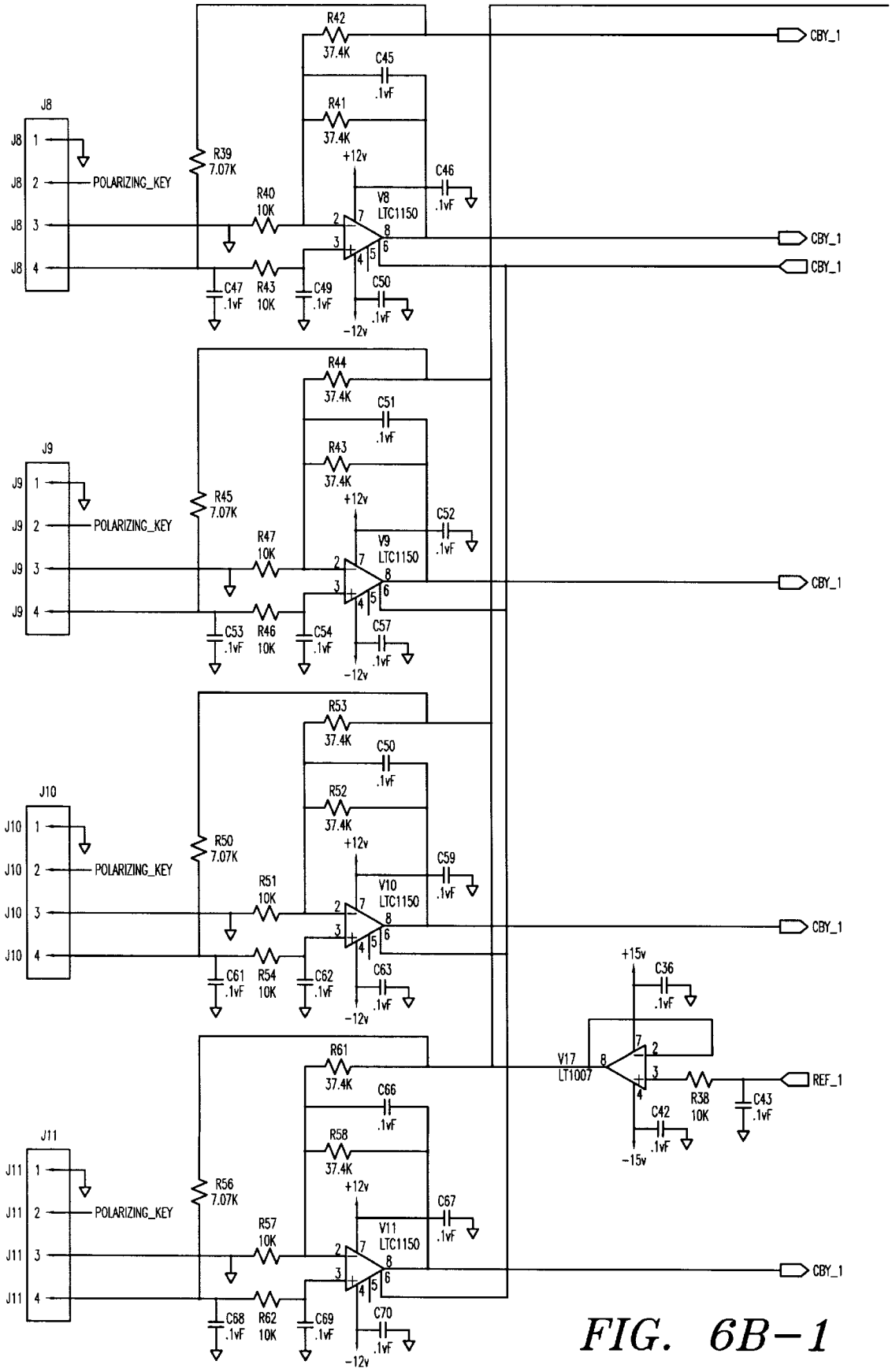

J8
J8 1
J8 2
J8 3
J8 4
POLARIZING_KEY
R39 7.07K
R40 10K
R41 37.4K
R42 37.4K
C45 .1vF
C46 .1vF
V8 LTC1150
+12v
-12v
C47 .1vF
R43 10K
C49 .1vF
C50 .1vF
CBY_1
CBY_1
CBY_1
J9
J9 1
J9 2
J9 3
J9 4
POLARIZING_KEY
R45 7.07K
R47 10K
R43 37.4K
R44 37.4K
C51 .1vF
C52 .1vF
V9 LTC1150
+12v
-12v
C53 .1vF
R46 10K
C54 .1vF
C57 .1vF
CBY_1
J10
J10 1
J10 2
J10 3
J10 4
POLARIZING_KEY
R50 7.07K
R51 10K
R52 37.4K
R53 37.4K
C50 .1vF
C59 .1vF
V10 LTC1150
+12v
-12v
C61 .1vF
R54 10K
C62 .1vF
C63 .1vF
CBY_1
J11
J11 1
J11 2
J11 3
J11 4
POLARIZING_KEY
R56 7.07K
R57 10K
R58 37.4K
R61 37.4K
C66 .1vF
C67 .1vF
V11 LTC1150
+12v
-12v
C68 .1vF
R62 10K
C69 .1vF
C70 .1vF
CBY_1
+15v
C36 .1vF
V17 LT1007
R38 10K
C43 .1vF
C42 .1vF
-15v
REF_1

FIG. 6B-1

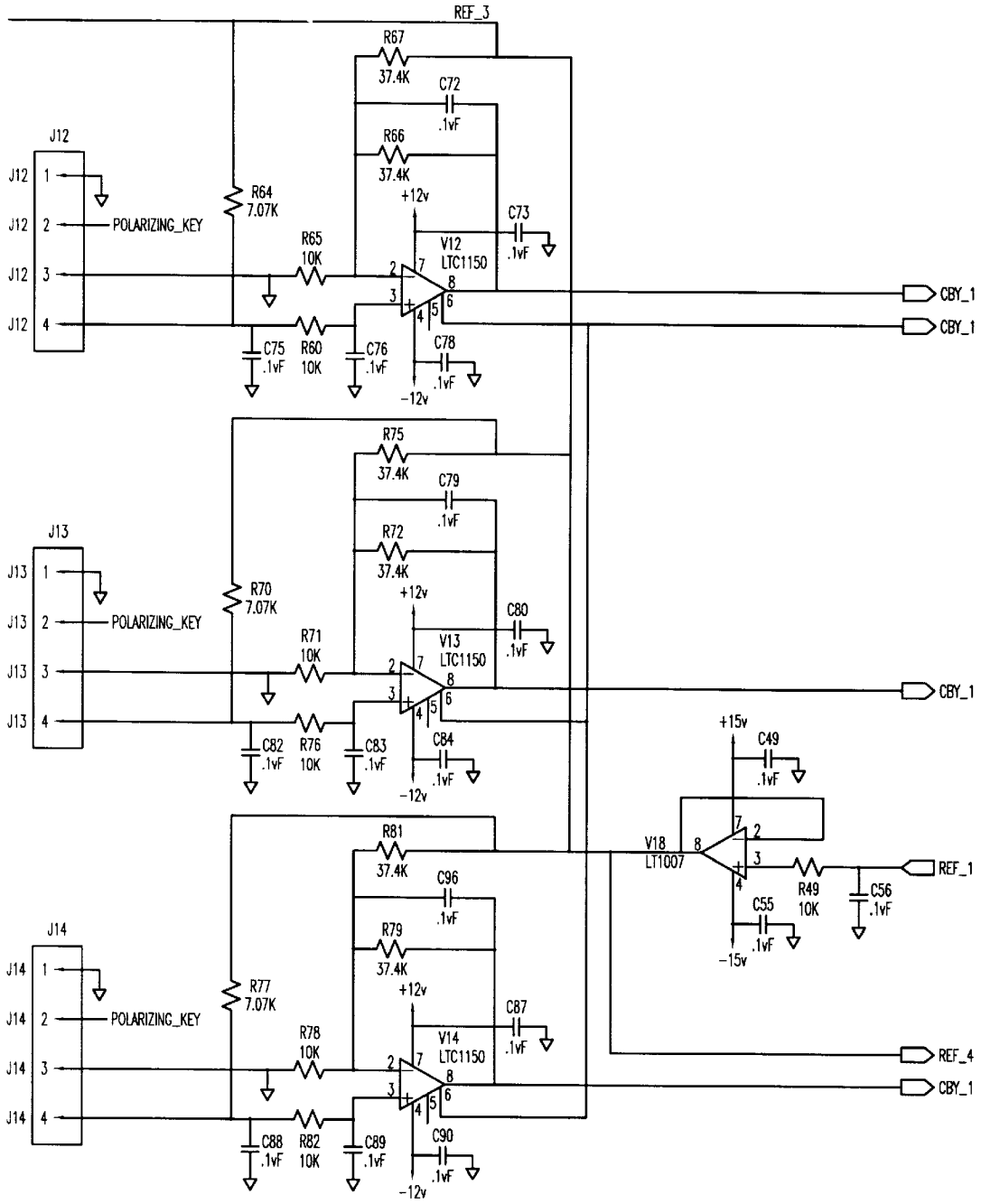


FIG. 6B-2

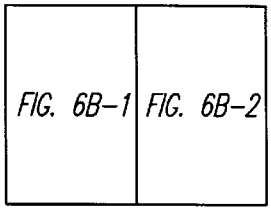


FIG. 6B

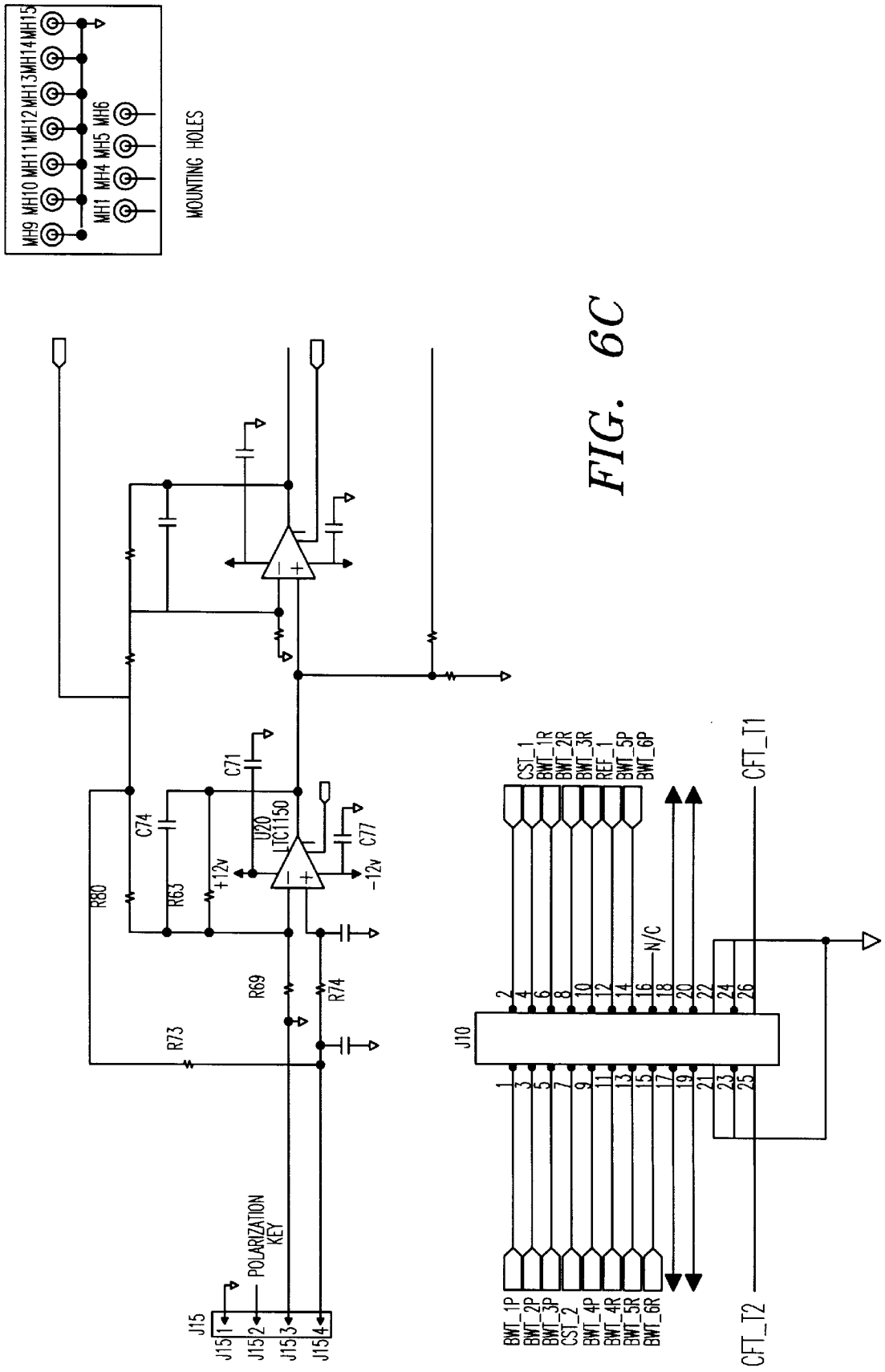

MH9 MH10 MH11 MH12 MH13 MH14 MH15
MH1 MH4 MH5 MH6
MOUNTING HOLES
C71
C74
C77
U20
LTC1150
+12v
-12v
R80
R63
R69
R74
R73
J15
POLARIZATION KEY
J15 1
J15 2
J15 3
J15 4
J10
CST_1
BWT_1R
BWT_2R
BWT_3R
REF_1
BWT_5P
BWT_6P
N/C
BWT_1P
BWT_2P
BWT_3P
CST_2
BWT_4P
BWT_4R
BWT_5R
BWT_6R
CFT_T1
CFT_T2

FIG. 6C

TEMPERATURE CONTROL OF HIGH TEMPERATURE SUPERCONDUCTING THIN FILM FILTER SUBSYSTEMS

This application is a continuation of U.S. Ser. No. 09/204,897 Dec. 3, 1998, now U.S. Pat No. 6,098,409.

BACKGROUND OF THE INVENTION

The field of the present invention is temperature control systems and, more particularly, systems and methods for effecting temperature control of high temperature superconducting thin film filter sub-systems.

Recently, substantial attention has been devoted to the development of high temperature superconducting radio frequency (RF) filters for use in, for example, cellular telecommunications systems. However, such filters are extremely temperature sensitive. By their very nature, high temperature superconducting (HTSC) materials are temperature dependent. At temperatures above their "transition temperatures," the materials behave like an insulator, and at temperatures below the transition temperature, the materials become superconducting.

Further, when a HTSC film is fabricated into a RF filter, temperature fluctuations stemming from kinetic inductance of the filter may have a substantial effect upon the operation of the filter and, in particular, upon the center-frequency of the filter. Similarly, fluctuations in temperature may have a substantial impact upon certain non-linear behavior characteristics of HTSC thin film filters. While the non-linear behavior characteristics of a HTSC thin film filter may have a relatively mild effect upon filter operation at temperatures below the transition temperature, the same cannot be said for the kinetic inductance effect. Further, as the temperature of operation of a HTSC thin film filter approaches, for example, the transition temperature of the filter, relatively minor fluctuations in the operating temperature can have very significant effects upon filter operation. Stated somewhat differently, as HTSC thin film filter systems are operated closer and closer to their respective transition temperatures, more and more care must be taken to control the temperature of the operating environment. Thus, it will be appreciated that HTSC thin film filter systems must be maintained at stable operating temperatures if proper operation of the systems is to be maintained. This is particularly so where HTSC filters are to be operated at or near their respective transition temperatures.

Those skilled in the art also will appreciate that increased temperature stability generally is required when more "narrow-band" filters are utilized within a HTSC filter system. The reason for this is that relatively small changes in operating temperatures (e.g., +/−1° K.) may have a substantial impact upon the range of filter operation, particularly if a filter is operated at or near its transition temperature. Indeed, such changes in operating temperature may cause the center frequency of a HTSC filter to vary by as much as 100 MHz.

Now, because maximum advantage may be obtained through the use of HTSC thin film filters when the filters are operated in a narrow-band mode at approximately the transition temperature, those skilled in the art will appreciate that it is highly desirable, if not essential, to maintain very precise control of the operating temperatures of HTSC thin film filter systems.

Those skilled in the art also will appreciate that, when multiple HTSC filters are disposed, for example, within the dewar of a cryocooler, and the cryocooler is mounted, for example, on a telecommunications tower, substantial temperature control issues may arise. Simply put, a tower-mounted cryocooler will need to provide more lift (i.e., more "cold") on a hot afternoon than would be required on a cold night. Further, as the ambient temperature of the environment within which a HTSC filter system is mounted varies, temperature gradients will result between the system cold source (i.e., the cold finger of the system cryocooler) and the cold stage or location where the HTSC filters are located.

Traditionally, the above-described problems have been solved by (1) operating a HTSC filter at temperatures well below the transition temperature of the filter; (2) controlling the temperature of a HTSC filter by adding heat to a so-called cold plate with some thermal conduction to a cooler; (3) over-designing a HTSC filter system so that required temperature specifications may be met even in the presence of substantial fluctuations in, for example, ambient temperature; (4) making a cold plate or HTSC filter mount very thick so as to reduce temperature gradients across the mount; and/or (5) providing for in-situ tuning of a HTSC filter system.

Those skilled in the art will appreciate that each of the above-described options represents only a partial solution to the HTSC filter temperature control issue, and that each option has inherent disadvantages. For example, option (1) represents a tradeoff between filter temperature stability and cooler size. In short, the use of a larger cooler may provide lower "cold-finger" temperatures and increased lift, but may result in higher power consumption and system heat generation. Option (2) represents a similar tradeoff, but in that instance, the issue that must be addressed is the addition of a new heat load to the system. Finally, those skilled in the art will appreciate that options (3) and (5), while potentially effective, are subject not only to economic limits, but also to performance limits.

In view of the foregoing, it is believed that those of ordinary skill in the art would find an improved temperature control system for use with HTSC thin film filter systems to be quite useful.

SUMMARY OF THE INVENTION

The present invention is directed to an improved temperature control system for HTSC thin film filter systems and to related temperature control methodologies.

In one particularly innovative aspect, the present invention is directed to the use of a multi-stage temperature controller within a HTSC thin film filter system. In one presently preferred embodiment, the multi-stage temperature controller comprises a first temperature control loop, or "inner loop," and a second temperature control loop, or "outer loop." The inner loop monitors a cold finger temperature of a cryocooler and provides for efficient and stable control of the cryocooler in response to an internal reference signal or voltage. The outer loop monitors a cold stage temperature (i.e., a temperature within the general vicinity of a HTSC filter), and provides (as an output) the internal reference voltage to the inner loop. Thus, it will be appreciated that a temperature control system in accordance with the present invention has the ability to quickly and efficiently respond to temperature gradients that may exist between a cold stage of a filter system and, for example, the cold finger of an associated cryocooler. As explained above, such gradients may result, for example, from temporal, seasonal and/or weather-induced changes in ambient temperature, and it is quite important that a tower-mounted HTSC filter system have the ability to accommodate such ambient temperature changes.

It also will be appreciated that, by using a multi-staged control system in accordance with the present invention, it is possible to precisely monitor the temperature within the immediate vicinity of one or more HTSC filters or filter stages without sacrificing the stability of any relevant control loop. This, it is believed, represents a significant improvement over conventional HTSC filter temperature control systems.

In other preferred embodiments, a temperature control system in accordance with the present invention may further include a filter loop of the type described in co-pending U.S. patent application Ser. No. 08/369,004, which is entitled "Temperature Controlling Cryogenic Package System" and is incorporated herein by reference. The filter loop may be used to control the current provided to one or more resistors located within the immediate vicinity of the HTSC filters and, therefore, to fine tune the temperatures of the various filters within a system. Such a loop also may be used to adjust with substantial accuracy the center frequency of the filters.

Finally, in still further embodiments, a plurality of HTSC thin film filter package assemblies may be mounted to an innovative, symmetrically shaped heat-sink in accordance with the present invention. The use of such a heat-sink allows for the use of a single pair of sensors (e.g., a cold finger temperature sensor and a cold stage temperature sensor) to be used to control with fairly high precision the operating temperatures of several HTSC filter package assemblies. The use of such a heat-sink also allows a temperature sensor provided, for example, on one cold stage to predict with a fairly high degree of accuracy the temperature of another cold stage within a HTSC filter system.

Accordingly, it is an object of the present invention to provide an improved temperature control system for use with, for example, HTSC thin film filters, and to provide improved methods for regulating the temperatures of such filters.

Other objects and features of the present invention will become apparent from consideration of the following description taken in conjunction with the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is an isometric view of a heat-sink for use with a cryocooler in a temperature control system in accordance with the present invention.

FIG. **2** is a block diagram illustrating the components of a multi-stage temperature controller in accordance with one form of the present invention.

FIG. **3** is a block diagram illustrating the components of a multi-stage temperature controller in accordance with a preferred form of the present invention.

FIG. **4** is an illustration of a HTSC filter carrier package that may be used in accordance with the present invention.

FIGS. **5**(A)–(H) and **6**(A)–(C) comprise circuit schematics for a mother board and daughter board, respectively, of a temperature controller in accordance with a preferred form of the present invention.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

Turning now to the drawings, FIG. **1** provides an illustration of a heat-sink **10** that may support a plurality of HTSC thin film filter substrates (not shown) in accordance with the present invention. The heat-sink **10** is generally symmetric in shape and has central cavity **12** for mating with a cold finger (not shown) of, for example, a Stirling cycle cryocooler. In addition, the heat-sink **10** has a plurality of T-shaped radially extending extremities **14** that provide mounts for a corresponding number of HTSC filter assembly carrier packages (not shown). It will be noted that a cold finger temperature sensor **16** may be affixed to a main body portion **18** of the heat-sink **10**, and that one or more cold stage temperature sensor(s) **20** may be affixed to a central portion of an external surface **22** of the T-shaped extremities **14**. Thus, it will be appreciated that the cold finger temperature sensor **16** may be used to monitor with fairly close approximation a temperature of a cold finger (not shown) of an associated cryocooler, and the cold stage temperature sensor(s) **20** may be used to monitor with fairly close approximation the temperatures of any HTSC filter assembly carrier packages (not shown) that are mounted to the heat-sink **10**. A cryocooler of the type that might be used in accordance with the present invention is described in co-pending U.S. patent application Ser. No. 09/175,924, which is entitled "Cryocooler Motor with Split Return Iron" and is hereby incorporated by reference.

Now, turning to FIGS. **2**, in one presently preferred form, a temperature controller **30** for a HTSC thin film filter system including, for example, a Stirling cycle cryocooler **40** may comprise a multi-staged system including an inner loop controller **32** and an outer loop controller **34**. The inner loop controller **32** receives, as a first input, an output signal **33** from the cold finger temperature sensor **16**. As a second input, the inner loop controller **32** receives a signal comprising the output **36** of the outer loop controller **34**. Thus, it will be appreciated that the output **36** of the outer loop controller **34** may be viewed as a reference signal to which the signal **33** from the cold finger temperature sensor **16** is compared. The output **38** of the inner loop controller **32** is used to control the amount of power that is provided to the Stirling cycle cryocooler **40**. Accordingly, it will be appreciated that an inner loop of a control system in accordance with the present invention may comprise a fairly fast, simple and stable loop for controlling to a first order the temperature of the cold finger (not shown) of the cryocooler **40**, and that the cold finger temperature of the cryocooler **40** may be dictated by the output **36** of the outer loop controller **34**.

In contrast, the outer loop controller **34** has, as a first input, a reference signal **42** and, as a second input, an output signal **44** received from the cold stage temperature sensor **20**. Thus, the signal **44** received from the cold stage temperature sensor **20** may be compared to the reference signal **42** by the outer loop controller **34**, and the outer loop controller **34**, in turn, may be used to drive, or set the reference temperature for, the inner loop of the control system. In this fashion, temperature gradients within the heat-sink **10**, that may result, for example, from temporal, seasonal or weather-induced changes in ambient temperature, may be easily accommodated by the control system **30**.

Turning now to FIG. **3**, in a particularly preferred form, a temperature control system **50** in accordance with the present invention is microprocessor based. Moreover, a microcontroller **52** and related EPROM **54**, interface logic **56**, and analog-to-digital (A/D) converter **58** are utilized to monitor selected points μp within the control system **50**. Those skilled in the art will appreciate that the temperature control system **50** may be implemented using either analog or digital circuits and, therefore, that the specific circuitry utilized to implement the control system **50** is not described in detail herein. Rather, the basic functionality of the control

system **50** is described herein, and to the extent that specific details regarding the implementation of the control system **50** may be useful to those skilled in the art, detailed circuit schematics of a multi-staged control system in accordance with the present invention are submitted herewith as FIGS. **5**(A)–(H) and **6**(A)–(C).

Like the basic system **30** illustrated within FIG. **2**, the control system **50** preferably includes an inner control loop and an outer control loop. However, in the case of the control system **50**, the inner loop is somewhat more refined. For example, the inner loop of the control system **50** may comprise, in addition to the inner loop controller **32** and cold finger temperature sensor **16**, a thermistor sensor **60** and related processor **62**, a knock sensor **64** and related processor **66**, a limiter circuit **68**, and a multiplier circuit **70**. Further, it will be appreciated that, while the inner loop controller **32** may comprise any type of proportional controller, it is presently preferred that the inner loop controller comprise a proportional and derivative (PD) controller.

The knock sensor **64** may comprise, for example, a PZT crystal based sensor (or microphone), a back EMF sensor circuit or current analyzer circuit that may be used to detect overstrokes of the cryocooler **40**. Moreover, it will be appreciated that more power may be provided to a Stirling cycle cryocooler as the temperature of the cold finger (not shown) of the cryocooler decreases. Stated somewhat differently, when a Stirling cycle cryocooler initially is turned "on" only a limited amount of power may be provided to the cryocooler to drive the cryocooler. However, as the temperature of the cold finger (not shown) of the cryocooler begins to drop, additional power may be provided to the cryocooler. Thus, the knock sensor **66**, when used in conjunction with the limiter circuit **68**, functions to ensure that the cryocooler **40** is not overdriven, for example, at start-up.

Like the knock sensor **66**, the thermistor sensor **60** and limiter circuit **68** also function to ensure that the cryocooler **40** is not overdriven. However, the thermistor sensor **60** is used to monitor the temperature of the cryocooler **40** and to limit the power that is provided to the cryocooler **40** when, for example, the cryocooler **40** becomes too hot.

The multiplier circuit **70** preferably is responsive to the output signal **72** of the limiter circuit **68** and amplitude modulates a 60 Hz signal that is used to drive the cryocooler **40**.

Finally, it will be appreciated that the inner loop controller **32** receives as inputs the output **36** of the outer loop controller **34**, a reference signal **74**, and a signal comprising the output **33** of the cold finger temperature sensor **16**. However, the output **38** of the inner loop controller **32** preferably is primarily derived from the output **36** of the outer loop controller **34**.

Thus, it will be seen that the output of the inner loop controller **32**, as modified by the limiter circuit **68** and provided to the multiplier circuit **70**, may be used to drive, for example, a Stirling cycle cryocooler **40** causing the cryocooler **40** to provide variable lift (i.e., a varying source of "cold"). It will further be appreciated that the inner loop provides a comparatively fast, simple and stable loop that drives the cold finger temperature of a cryocooler **40** to a temperature dictated by the outer loop controller **34**.

Turning now to the function of the outer loop controller **34**, that controller preferably comprises a proportional integral derivative (PID) controller, and has as its inputs the reference signal **74** and a signal comprising the output **44** of the cold stage temperature sensor **20**. The outer loop controller **34** strives to provide "zero error" between the cold stage sensor output signal **44** and the reference signal **74**. The reference signal **74** preferably is set by a precision voltage source **78** that is adjustable via software and is used to determine the ultimate cold stage operating temperature. As mentioned above, a portion of the reference signal **74** is provided as an input to the inner loop controller **32** in an effort to improve a dynamic range of the control system **50** independent of the outer loop set point.

It will be appreciated that, through the use of a dual control loop configuration in accordance with the present invention, it is possible to provide a control system that is stable, comparatively quick, and quite accurate over a wide variety of operating conditions. Further, because of the relative speed of the inner loop, the inner loop preferably tends to move the "dominant pole" of the control system **50** to a higher frequency. This, in turn, allows for thermal accuracy purposes optimization of the placement of the cold stage temperature sensor **20**, and also provides for improved dynamic stability of the control system **50**.

The configuration of the heat-sink **10** also enhances system function and, if desired, may allow for a single cold stage temperature sensor **20** to be utilized to monitor within a very close approximation the temperatures of multiple HTSC filter assembly carrier packages.

As has been explained to some extent above, one principal advantage that is achieved through the use of a control system **50** in accordance with the present invention is that the control system **50** has the ability to respond to temperature gradients that may exist between the cold finger and cold stages of a HTSC filter system enclosure. Moreover, the utilization of a temperature control system **50** in accordance with the present invention makes it possible to achieve thermal stability of a large thermal mass, while at the same time maintaining dynamic stability of the control system and using comparatively inexpensive parts.

Finally, a temperature control system in accordance with the present invention may include a "filter loop," of the type described within co-pending U.S. patent application Ser. No. 08/369,004, which is entitled "Temperature Controlling Cryogenic Package Systems." Moreover, each HTSC filter (not shown) may be attached to a carrier (shown in FIG. **4**) that is thermally matched to the HTSC substrate. The carrier packages **90** shown, for example, in FIG. **4** are mounted on the heat-sink **10**, and preferably are thermally isolated from the heat-sink **10** through the use of glass pads **92** that are provided on the mounting edges **94** of the carrier packages **90**. Thick film resistors **96** and a temperature sensor **98** also are provided on the carrier packages **90**, and are used to regulate with very high precision the temperatures of the carrier packages **90**. Preferably, a PID controller **100** is used to regulate the voltage provided across the resistors **96** of the respective carrier packages **90** and, thus, to regulate (or fine tune) the temperatures of the respective carrier packages **90** by modulating the amount of heat provided thereto.

It will be appreciated that, because the multi-staged temperature control system **50** provides for highly accurate control of the temperature(s) of the cold stage(s), the temperature modulation provided by the filter loops need only accommodate a fairly limited range (i.e., on the order of 1° K.).

Preferably, the carrier packages **90** are designed such that the masses of the carrier packages **90** and HTSC filters (not shown) carried thereon are quite small, and such that the thermal impedance between the carrier packages **90** and the

heat-sink **10** is relatively high. The net result is a filter temperature control loop which has a very fast response time and has the ability to "track out" any temperature changes that may appear on the cold stages. The filter loops also may be used to tune and/or adjust the center frequencies of the HTSC filters provided on the various carrier packages **90**.

In view of the foregoing, it will be appreciated that the use of a temperature control system in accordance with the present invention provides for very precise cold stage temperature regulation, while at the same time providing and/or allowing for reduced system power consumption, high stability within all control loops, elevated filter operating temperatures (i.e., operation at temperatures approaching HTSC filter transition temperatures), and the use of very "sharp" HTSC filters.

While the invention is susceptible to various modifications and alternative forms, specific examples thereof have been shown in the drawings and are herein described in detail. It should be understood, however, that the invention is not to be limited to the particular forms or methods disclosed, but to the contrary, the invention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the appended claims.

What is claimed is:

1. A temperature controller for use with HTSC thin film filter systems, said temperature controller comprising:

an inner control loop for controlling a cold finger temperature of a cryocooler, the inner control loop comprising an inner loop controller and a cold finger temperature sensor, the inner loop controller receiving as inputs a signal from the cold finger temperature sensor, a reference signal, and a signal comprising the output of an outer loop controller, wherein the inner loop controller includes an output signal used to control the amount of power used to drive the cryocooler; and

an outer control loop comprising an outer loop controller and a cold stage temperature sensor, the outer loop controller receiving as inputs a signal from the cold stage temperature sensor and said reference signal, wherein the outer loop controller outputs a signal to the inner loop controller.

2. A temperature controller according to claim **1**, wherein the inner loop controller is a proportional and derivative (PD) controller.

3. A temperature controller according to claim **1**, wherein the outer loop controller is a proportional integral derivative (PID) controller.

4. A temperature controller according to claim **1**, wherein the reference signal is set by a voltage source.

5. A temperature controller according to claim **4**, wherein the voltage source is adjustable via software.

6. A temperature controller according to claim **1**, wherein the output of the inner loop controller is input to a limiter circuit, the limiter circuit receiving a first signal from a thermistor sensor used to monitor the temperature of the cryocooler and a second signal from a knock sensor used to detect overstrokes of the cryocooler.

7. A temperature controller according to claim **6**, wherein the limiter circuit outputs a signal to a multiplier circuit used to drive the cryocooler.

8. A temperature controller according to claim **6**, wherein the knock sensor is a sensor selected from the group consisting of a PZT crystal based sensor, back EMF sensor circuit, and current analyzer circuit.

9. A temperature controller according to claim **1**, wherein the cryocooler is a Stirling cycle cryocooler.

* * * * *

UNITED STATES PATENT AND TRADEMARK OFFICE

CERTIFICATE OF CORRECTION

PATENT NO. : 6,256,999 B1
DATED : July 10, 2001
INVENTOR(S) : Chase

It is certified that error appears in the above-identified patent and that said Letters Patent is hereby corrected as shown below:

Column 1,
Line 52, change "K.)" to -- K) --

Column 4,
Line 23, change "FIGS." to -- FIG. --.

Column 5,
Lines 33 and 37, change "sensor 66" to -- sensor 64 --.

Signed and Sealed this

Tenth Day of September, 2002

Attest:

Attesting Officer

JAMES E. ROGAN
Director of the United States Patent and Trademark Office